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Using machine learning to estimate corporate carbon emissions: Is it all just hot air?

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Abstract

Introduction: Corporate greenhouse gas emissions predominantly occur through the value chain, classified as "Scope 3" emissions. These emissions arise indirectly as a consequence of the firm's activities, for example through supply chain operations and product use, making them difficult to measure and control. Moreover, accurately estimating Scope 3 emissions is complex but resource-intensive as they fall outside of a firm's direct oversight. While machine learning has been explored as a potential solution, previous efforts have failed to significantly outperform simpler estimation methods, leaving predictive accuracy low. This limitation underscores the need for more effective approaches to improve emission estimates, which are critical for both corporate sustainability goals and regulatory compliance.

Objective: The objective of this paper is to investigate the challenges that machine learning methods face in accurately estimating Scope 3 emissions. Particular focus is given to understanding how data-related issues impact model performance, offering insights into the connection between data quality, feature selection, and predictive accuracy.

Methods: To address this objective, I employ an empirical approach that examines four critical dimensions of the modelling process: target variable selection, feature engineering, sample composition, and model choice. By systematic experimentation with different combinations of these elements, I identify both successful strategies and persistent challenges. Additionally, I conduct three complementary analyses to deepen my understanding of performance limitations. First, I examine some outliers produced by the even most accurate model. Second, I apply a materiality filter to highlight the trade-off between data quality and data coverage. Finally, I explore the constraints of category-level modelling, emphasising the need to aggregate Scope 3 emissions data to a sufficient level for more reliable predictions.

Results: Scope 3 emissions are best estimated by: using emission intensities (that is, emissions divided by revenue); a core set of features; and (where sufficient data exists) by focusing solely on material disclosures. Each of these choices significantly improves predictive performance compared to the alternatives. In contrast to these unambiguous choices, no single model architecture clearly outperforms the others for all target variables, with machine learning models offering performance which is comparable to simpler models. Nevertheless, by filtering the data using material disclosures, my best Scope 3 model achieves superior predictive accuracy over the best model for Scopes 1 and 2. Furthermore, with respect to predictive accuracy, these results represent a substantial improvement over previous machine learning attempts to estimate Scope 3 emissions.

Conclusion: Overall, my findings suggest that improving Scope 3 emission estimates depends primarily on addressing limitations in data coverage and quality rather than

refining model architecture. While enhancing data inputs leads to significant gains in predictive accuracy, the comparable performance of machine learning models to simpler models indicates that the core challenge in Scope 3 modelling is not complex interactions or non-linear relationships, which machine learning is typically suited to address. This highlights the need for improved data practices over more advanced modeling techniques when tackling Scope 3 estimation.

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Both a hearty thanks and apology are owed to my beautiful wife, Sharon, who has borne much of the cost that Imperial College London exacts. I also record my thanks to Dr Rob Craven, my MSc co-ordinator, for being wellspring of empathy and good cheer.

I am also greatly obliged to my parents, David and Jane, and 박금일 목사님 and 손명은 사모님 for their emotional, spiritual and financial support throughout the year.

I dedicate this thesis to my God, who by his providence has taught me that “a man’s heart plans his course, but the LORD determines his steps.”

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Chapter 1

Introduction

Almost every country in the world has committed to the Paris Agreement, a legally binding international treaty that aims to limit the global temperature increase above pre-industrial levels to 1.5 degrees Celsius: to achieve this, greenhouse gas (GHG) emissions must peak before 2025 and decline 43% by 2030 [74]. Therefore, at the time of writing, we should have reached peak GHG emissions and must expect steep reductions to follow, especially in carbon dioxide which comprises most of the GHG mix [57].

For these ambitious targets to be achieved, firms must play a part to reduce their carbon emissions. As "what gets measured, gets managed", a necessary first step is for firms to account and report their GHG emissions. This need motivated the GHG Protocol Initiative, an international accounting and reporting standard for business [75]. To assist firms, the Protocol provides a framework that divides emissions into three categories: Scope 1 (S1) encompasses a firm's direct emissions; Scope 2 (S2) are GHG emissions from purchased energy; and Scope 3 (S3) are all other indirect emissions from the value chain.

With the GHG Protocol steadily being adopted, over time the availability of GHG emissions data has improved. Nevertheless, significant problems remain with the coverage and accuracy of S3 data [52]. Startlingly, while on average over 80% of a firm's carbon footprint is from S3 emissions, only around half of firms report some S3 emissions data, and only 20% of firms report material S3 emissions [25]. To try to fill these gaps, attempts have been made to use machine learning (ML) techniques [49, 48]. However, compared to naive baselines such as the sector median, these state-of-the-art approaches have shown only marginal improvements. This is also reflected by the practice of third-party providers, who often use simple models to impute missing S3 emissions.

The goal for this project is to understand why ML approaches struggle to improve on simple baselines and statistical methods for estimating S3 emissions. Specifically, I investigate how issues like data quality, feature selection, and sample composition influence model performance, offering insights into why machine learning fails to deliver a breakthrough in predictive accuracy. I begin in Chapter 2 with providing

the necessary background to the domain of S3 data, with a focus on data quality and coverage. After describing my sample construction in Chapter 3, in Chapter 4 I lay out my modelling strategy. This is a systematic, experimental approach to find what does and does not work. After getting the best results I can, presented in Chapter 5, in Chapter 6 I consider where the models struggle and what alternatives there are. Using the lessons drawn from this, in Chapter 7 I offer some suggestions for promising avenues of applied ML work in the S3 domain. Chapter 8 concludes.

1.1 Legal, social, and ethical requirements

To meet the requirements and expectations of this project, I have considered all of the potential legal, social, ethical and professional issues as highlighted in the Department of Computing's checklist. As my project concerns estimating corporate carbon emissions, there are no concerns regarding any human participants, animals, personal data, developing countries, environmental safety, dual-use, misuse or other ethical issues.

Regarding data disclosure issues, I have been careful to avoid publishing any data to my GitHub repository that might result in licensed data leakage. Furthermore, in consultation with LSEG professionals, I have also taken steps to avoid making anything public which could be considered sensitive. For example, this is why in Section 6.1, I refer to specific firms using generic labels (Firm A, Firm B, Firm C) in code-generated plots, so that my code base avoids using identifiable firm names. Moreover, this report (unless processed further) will remain private and, combined with the LSEG-Imperial data sharing agreement, this should ensure that there are no problems arising from using licensed data.

Finally, there are important legal considerations for ML in a commercial context. Specifically, integrating my code into a commercial solution would require careful attention to issues surrounding explainability and transparency. While the logic of simpler models like decision trees and nearest neighbours is relatively intuitive, the construction of large models using high-dimensional data can significantly reduce their interpretability. This is particularly true of models such as neural networks, which rely on hidden layers to process abstract representations of data, making it difficult to audit the decision-making process.[1]

The European Union's Artificial Intelligence Act (AI Act) places strong emphasis on the need for adequate explainability in AI systems, especially when these systems impact high-stakes decisions such as financial services and capital allocation. Article 13 of the AI Act requires that "high-risk AI systems shall be designed and developed in such a way as to ensure that their operation is sufficiently transparent to enable deployers to interpret a system's output and use it appropriately." [11] Failing to

[1]This is further complicated by standard feature attribution methods not being valid in a time-series context. See the discussion in Section 4.2.

meet these standards may result in regulatory non-compliance and consequent legal liabilities.

Moreover, the growing regulatory landscape in data privacy and the responsible use of data also imposes restrictions. For instance, using forecast emissions data in a ML model could trigger issues related to data provenance and the responsibility of ensuring that published data is accurate and used within the bounds of intellectual property law. Both the General Data Protection Regulation (GDPR) in Europe and similar frameworks globally emphasise the need for clear accountability in how data is collected, processed, and shared, especially when it is used to predict outcomes with significant social or environmental impacts.

Chapter 2

Background

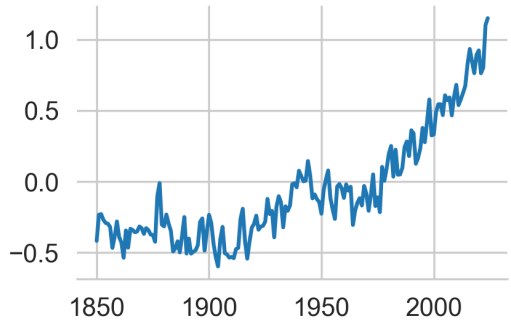
This chapter introduces the necessary background into S3 GHG emissions reporting, standards and data. I begin with the historical and policy context in Sections 2.1 and 2.2 respectively. This serves as a springboard to consider how firms are expected to report their S3 emissions in Section 2.3. Next, I cover the data quality issues inherent to this reporting process in Section 2.4. Finally, I conclude by reviewing various approaches in the literature to tackling these issues in Section 2.5.

2.1 Historical context

Beginning in the 1800s, naturalists were puzzled to find signs of glacial activity in places too warm for ice sheets to form [77]. Then, as the century progressed, some scientists discovered that certain gases such as carbon dioxide caused a "greenhouse effect" in their experiments [17]; therefore, they hypothesised that these GHGs might affect the planet's temperature through trapping heat in the atmosphere [1]. Accordingly, they found it plausible that past climates could have been very different from our own [22].

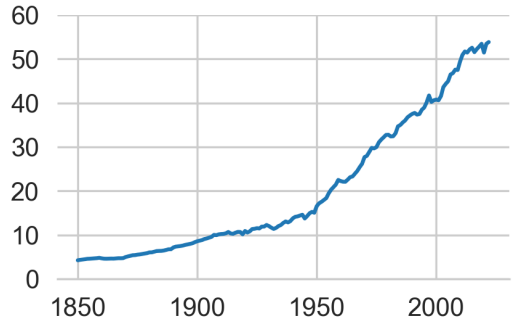
Consistent with this theory, in the twentieth century scientists observed rising global surface temperatures and steep increases in the concentrations of GHGs [58]. Thus, today's mainstream scientific consensus is that the earth's recent climate change is mostly driven by human activity: as economic development increased combustion of fossil fuels, this has raised GHG levels in the atmosphere resulting in higher global temperatures [35, 68].

Accompanying the growing weight of scientific evidence was political conviction that something must be done. Consequently, landmark international treaties were signed such as the 1997 Kyoto Protocol [73] and the 2015 Paris Agreement [72] which legally required countries to reduce their GHG emissions. Building on this scientific and political momentum around climate change, growing popular concern for the environment further catalysed domestic support for national targets such as Net Zero [55].



(a) Average global temperature anomaly

Global average land-sea temperature anomaly relative to the 1961-1990 average temperature, in degrees Celsius.



(b) Global GHG emissions

GHG emissions include carbon dioxide, methane and nitrous oxide from all sources, including land-use change. They are measured in billions of tonnes of carbon dioxide equivalents over a 100-year timescale.

Figure 2.1: Global average temperature anomalies and GHG emissions, [57]

2.2 Policy context

Accurate GHG emissions data are vital for meeting legal obligations and reduction targets, but this is not their only use. As this data also measures the demand for GHG emissions, price discovery is possible for emissions trading markets, meaning that the externalities are appropriately priced and so excess emissions are prevented. As firms disclose their carbon footprints, investors may also better hedge transition risk and follow their ethical preferences. Finally, as GHG emissions are ultimately an unwanted source of inefficiency, good GHG accounting helps identify opportunities for reducing waste.

In light of this, policy makers recognised the need for an international standard for corporate GHG accounting and reporting. Therefore, in 1998, the GHG Protocol Initiative was launched with a mission to develop and promote the adoption of GHG reporting standards for businesses [75]. Since then, the Protocol has increasingly been incorporated into global reporting standards including the Global Reporting Initiative (GRI 305, [26]), the International Financial Reporting Standards (IFRS S2, [33]) and the Taskforce on Climate-related Financial Disclosures (TCFD, [67]). Through its adoption into legal frameworks and voluntary initiatives, the Protocol has become a popular framework for GHG emissions accounting.

The GHG Protocol Corporate Standard divides a company's emissions into direct and indirect emissions. Direct emissions come from sources that are owned or controlled by the company. On the other hand, while indirect emissions are a consequence of company activity, they occur at sources outside of its ownership or control. To

help delineate this further, three "scopes" are defined for accounting and reporting purposes. First, scope 1 (S1) emissions are direct emissions from sources which are owned or controlled by the company. Second, scope 2 (S2) emissions are indirect emissions from the generation of energy that is purchased, such as from electricity, steam, heat and cooling. Finally, scope 3 (S3) comprises all other indirect emissions that occur in the company's value chain. Together, these scopes encompass the total emissions footprint of a company summarised in Figure 2.2 below.

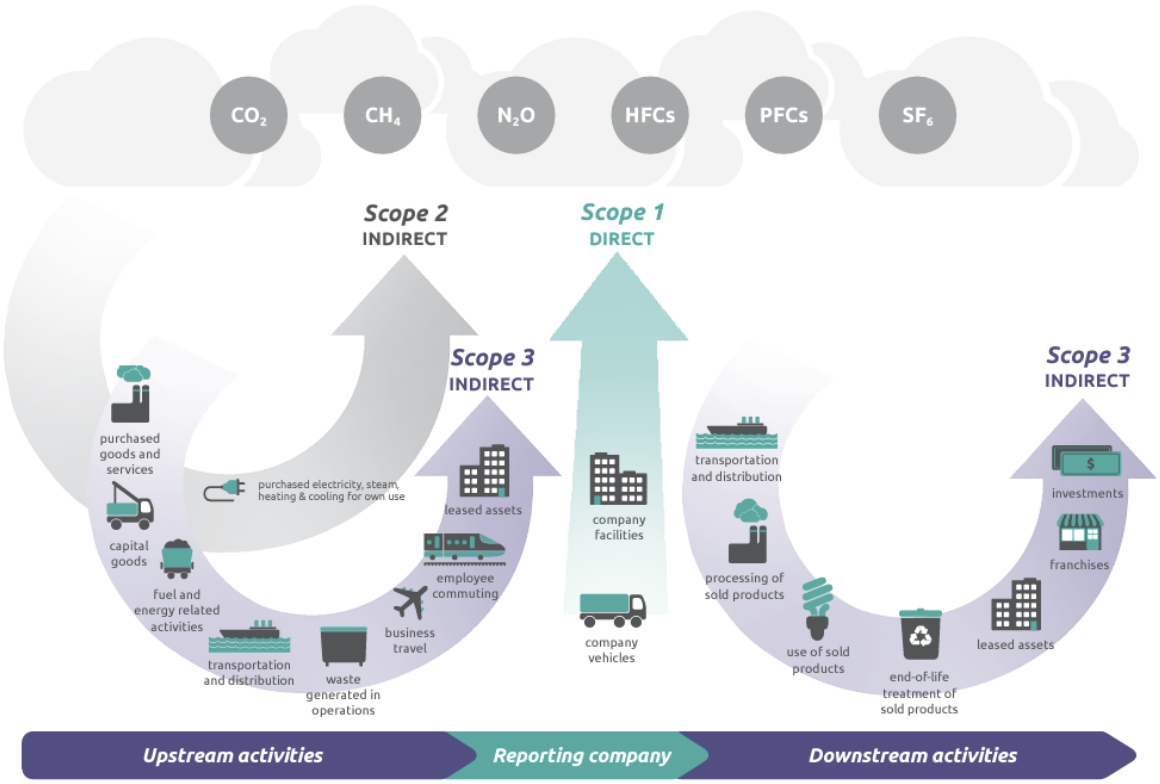


Figure 2.2: Overview of GHG Protocol scopes and emissions across the value chain, [76]

In accordance with the Protocol, many jurisdictions require (or will soon require) S1 and S2 disclosures for publicly listed firms, including in the EU [18], the US [59], the UK [70] and Japan [19]. Furthermore, although S3 reporting is currently optional in most of these key financial markets, the regulatory landscape is shifting away from voluntary efforts and towards mandatory S3 disclosures [25].

To illustrate these developments, I focus on how reporting requirements in the UK, one of the most progressive jurisdictions in climate reporting, have picked up over time. Beginning in 2013, listed companies were required to report their S1 and S2 emissions [70]. To bring in additional reporting requirements for listed companies, in 2018 the UK introduced the Streamlined Energy and Carbon Reporting (SECR) regime. More significantly, this legislation also brought in new requirements for large unquoted companies and limited liability partnerships to report their S1 and S2 emissions. In addition, it strongly encouraged S3 reporting from all companies, signalling the intended direction of travel [69]. Accordingly, the UK Government's

Table 2.1: Scope 3 Categories

	#	Category	Description
Upstream	1	Purchased goods and services	Extraction, production, and transportation of goods and services purchased or acquired by the reporting company
	2	Capital goods	Extraction, production, and transportation of capital goods purchased or acquired by the reporting company
	3	Fuel- and energy-related activities (not included in Scopes 1 or 2)	Extraction, production, and transportation of fuels and energy purchased or acquired by the reporting company not already accounted for in Scope 1 or Scope 2
	4	Upstream transportation and distribution	Transportation and distribution of products or services purchased by the reporting company between their tier 1 suppliers and own operations, including inbound and outbound logistics and inter-facility transport
	5	Waste generated in operations	Disposal and treatment of waste generated in the reporting company's operations
	6	Business travel	Transportation of employees for business-related activities
	7	Employee commuting	Transportation of employees between their homes and their workplaces
	8	Upstream leased assets	Operation of assets leased by the reporting company not included in Scope 1 or Scope 2
Downstream	9	Downstream transportation and distribution	Transportation and distribution of products sold by the company between the company's operations and the end consumer, including retail and storage
	10	Processing of sold products	Processing of intermediate products sold by downstream companies
	11	Use of sold products	End use of goods and services sold by the reporting company
	12	End-of-life treatment of sold products	Waste disposal and treatment of products sold by the reporting company at the end of their life
	13	Downstream leased assets	Operation of assets owned by the reporting company and leased to other entities not included in Scope 1 and Scope 2
	14	Franchises	Operation of franchises not included in Scope 1 and Scope 2
	15	Investments	Operation of investments (including equity and debt investments and project finance) not included in Scope 1 or Scope 2

Source: Adapted from The GHG Protocol Corporate Value Chain (Scope 3) Accounting and Reporting Standard [76]

stated ambition is to introduce mandatory S3 reporting across the entire economy by 2025 in line with the recommendations of the TCFD [71].

2.3 Scope 3 reporting

Having set out the origins and development of GHG emission targets, it remains to consider what is required for a firm to comply with the GHG Protocol's *Corporate Value Chain (Scope 3) Accounting and Reporting Standard* [76]. Possessing a solid understanding of this process will help illuminate the issues of data quality which are covered in Section 2.4.[1]

After the firm has divided its GHG emissions into direct and indirect, the first task is to set the S3 boundary; in short, this means accounting for all S3 emissions in the value chain and to disclose and justify any exclusions. This includes the gases contained in the Kyoto Protocol which are carbon dioxide ($C0_2$), methane (CH_4), nitrous oxide (N_20), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs) and lastly sulphur hexafluoride (SF_6). Furthermore, the company is required to account for these emissions in their respective S3 categories as listed in Table 2.1.

[1] S1 and S2 reporting is relatively straightforward as it follows from data within a firm's purview. Accordingly, S3 is where most effort on data coverage, consistency, quality and modelling is required [45], [52], [25].

These fifteen categories are designed to help firms organise, understand and report the diversity of S3 activities in their value chain. Though aiming for completeness, clearly in some cases accounting for every category will not be feasible. Therefore, the Protocol concedes that while some omissions might be permitted, "companies should ensure that the S3 inventory appropriately reflects the GHG emissions of the company ... In particular, companies should not exclude any activity that is expected to contribute significantly to the company's total S3 emissions" [76]. It remains for the company to judge what is "appropriate" and "signficant" to report.

After identifying and categorising the most likely sources of significant emissions in the value chain, the firm must then collect data for quantifying these emissions. There are two main methods to do this. First, where the firm has access to direct emissions data, they can simply use direct measurement. In practice, the second method of calculation is much more common, in which the firm uses activity data and emission factors to estimate emissions. In any case, the actual or estimated emissions data is then converted into a carbon dioxide equivalent ($C0_2e$) so that S3 emissions can be aggregated and compared regardless of GHG type. If any needed data remains missing, firms may use proxy data to fill gaps.

Using this data, the firm must next estimate their S3 emissions for each category. Importantly, companies may use a combination of calculation methods both within and across categories.[2] There are three calculation methods outlined in the GHG Protocol's S3 guidance and recognised by the literature [76, 27]. First, there is a process analysis (PA) approach which is bottom-up: it uses primary and secondary production process data and its associated carbon emissions. This has a high level of specificity and focus but may be costly or require data that is not possible to collect. Second, there is input-output analysis (IOA) which is top-down: it uses the input-output tables in the national accounts to allocate the carbon emissions of the economy to smaller units on a sectoral basis. This approach is simple to apply but by providing only averages in a population, it lacks nuance and it is difficult to show the results of reduction efforts. Third, there are hybrid approaches which combine PA and IOA to capture the strengths of both: under hybrid approaches, companies usually begin bottom-up and then fill any remaining gaps top-down [12].

Finally, to comply with the GHG Protocol S3 Standard [76], the firm must report their S3 emissions with the following requirements:

- Total scope emissions reported separately by each S3 category;
- For each S3 category, total GHG emissions reported in metric tonnes of $C0_2$ equivalent, excluding biogenic $C0_2$ emissions and independent of any GHG trades, such as purchases, sales, or transfers of offsets or allowances;
- A list of S3 categories and activities included in the inventory;

[2]Note, therefore, that the firm is expected to collect data first and then, based on what is available, apply the most appropriate method for each category, and not the other way around.

- A list of S3 categories or activities excluded from the inventory, and each with justification for their exclusion;
- Once a base year has been established: the year chosen as the S3 base year; the rationale for choosing the base year; the base year's recalculation policy; S3 emissions by category in the base year, consistent with the recalculation policy; and appropriate context for any significant emissions changes that triggered base year emissions recalculations;
- For each S3 category, any biogenic $C0_2$ emissions reported separately;
- For each S3 category, a description of the various types and sources of data, including activity data, emission factors and global warming potential (GWP) values, used to calculate emissions, and a description of the data quality of reported emissions data;
- For each S3 category, a description of the methodologies alongside allocation methods and assumptions used to calculate S3 emissions;
- For each S3 category, the percentage of emissions calculated using the data obtained from suppliers or other value chain partners.

These reporting requirements are clearly not trivial. Indeed, such is their nature and extent that some have concluded that they are practically unworkable in their current form [54]. The next Section explores the common problems that follow from these standards.

2.4 Data coverage and quality

Early studies using nascent S3 assessments found data that were largely incomplete and lacking rigour [16]. Unfortunately, even after a decade, a report by FTSE-Russell concluded that the data are still characterised by large gaps and poor quality [25]. This suggests that something in the data generating process (as outlined above in Section 2.3) could be driving these inadequate outcomes. This Section describes the main shortcomings of S3 data. Beginning with a review of challenges inherent in the data generating process, I subsequently highlight the problems this causes for analysis.

2.4.1 Issues with the data generating process

Klaaßen and Stoll [39] identify three causes of error in firms' S3 GHG emissions data: reporting inconsistency, boundary incompleteness and activity exclusion.

Reporting inconsistency

Firms may disclose their S3 GHG emissions through corporate channels including annual reports and corporate social responsibility reports (CSRs), to independent

third parties such as the CDP (formerly the Carbon Disclosure Project), or to their regulatory agencies. However, there is no need for these disclosures to be consistent with each other as they are written for different purposes and audiences. In line with the expectations of stakeholder theory, studies have found that on average, GHG emissions disclosed in corporate reports are lower than those reported to the CDP [15, 39]. As corporate reports are less prescriptive than the CDP questionnaire, it appears that firms use them to paint a better picture of their GHG emissions to the wider public by suppressing their bad outcomes and championing positive measures in so-called "green-washing". Notwithstanding this, different figures between reports do not necessarily imply fraudulent or dishonest disclosure: it may simply reflect the choice of valid but different methodologies [15]. In any case, the upshot is that even for a given firm in a given year which self-reports, there is potentially no consistent source of ground truth.

Boundary incompleteness

For each of the 15 S3 categories, the Protocol recommends a minimum boundary of what to include, and firms are then encouraged to choose the most appropriate calculation method depending on data quality and quantity. Calculation methods can be top-down (input-output based), bottom-up (process-based), or a hybrid of the two which begins bottom-up and then fills in the gaps top-down. For the most material categories, firms are expected to use primary data, i.e. data sourced directly from other firms. However, there is often limited data transparency across the value chain and so this option is commonly unavailable. Where secondary data such as industry reports and production volumes is also missing, boundary incompleteness follows. Some scholars such as Patchell [54] argue that this is an intractable problem due to the complexity of global supply chains and the prohibitive costs involved.[3]

Activity exclusion

Finally, even when firms report their S3 emissions, they may choose to omit certain relevant categories entirely. Despite the encouragements of standard setters to report S3 emissions that are considered 'material' (ISSB) [33], 'appropriate' (TCFD) [67], or 'significant' (GRI) [26], firms often report what is easier to measure (e.g. business travel) rather than what is important to their carbon footprint. For example, in their 2022 report, CDP found that only 36% of reporting firms disclosed their Purchased Goods and Services (Category 1) emissions, which are highly relevant for almost every sector [6, 25]. As this remains a persistent problem, FTSE Russell [25] has recommended that companies begin by reporting the two most material categories for their sector, which on average should encompass over 80% of their total S3 emissions. However, as will be shown Figure 2.3a and discussed further in Section 3.4, only around half of S3 disclosures satisfy the materiality requirement.

[3]This might explain why despite the US Securities and Exchange Commission (SEC) previously intending to introduce compulsory S3 disclosures, a move that some argued was a vital step for improving coverage [40], it has instead decided against this course of action following objections raised in industry consultations [59].

2.4.2 Consequential problems for data analysis

Missing data

The main reason for the paucity of data comes from the nature of S3 emissions: in contrast to S1 and S2, they are by definition emissions in the value chain beyond the firm's direct control or ownership, and therefore the data are often outside a firm's information set. Ideally, firms would be able to share data across the value chain, however there are significant barriers to doing this, including legal and regulatory challenges, interoperability and data privacy [63].

Accordingly, across various datasets, S3 coverage typically exists for only around half of firms.[4] Furthermore, as reporting remains voluntary and the propensity to report depends on firms' characteristics such as size, sector, and reporting jurisdiction, the missing data should be assumed to be 'missing not at random' (MNAR). However, current estimation models used by data providers often do not take this uncertainty into account, leading to misleading prediction intervals and inaccurate predictions [51]. In this way, missing data is closely related to biased data, the second problem in analysis.

Biased data

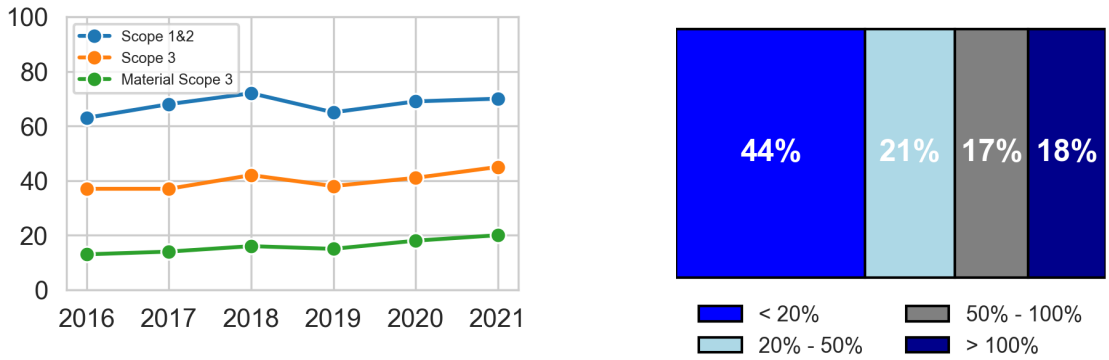
Whereas a firm which is non-reporting is conspicuous by its absence, a more subtle problem is when a firm under-reports due to boundary incompleteness or activity exclusion. Without scrutiny of a firm's methodology, the researcher might assume that the reported total S3 emissions is an accurate reflection of the firm's carbon footprint, rather than a reflection of what they choose to measure and report. This problem also further degrades the data generating process; as many firms under-report their emissions, firms which estimate their emissions using peer averages also under-report as a result [32]. Both problems of missing data (non-reporting) and biased data (under-reporting) are illustrated by Figure 2.3a, which shows that most firms don't report their S3 data (i.e. one minus the orange line), and for those that do, they mostly report immaterial figures (i.e. the gap between orange and green line).

Volatile data

The relative immaturity of S3 reporting is reflected in the high volatility of the data (see Figure 2.3b). As of 2021, over half of the disclosed values vary by over 20% year-on-year, and over a third vary by at least 50%. Although this is something that could improve over time as reporting practices evolve, the evidence suggests that the issue is still getting worse [25]. Moreover, even when FTSE-Russell considered S1 and S2 data which is much less volatile than S3, they found that irrespective of

[4] 45% of FTSE All-World constituents report S3 data as of 2021 [25]. For firms reporting to the CDP, 41% disclose emissions for one or more S3 category in 2022 [7]. Bloomberg and Refinitiv Eikon appear to have slightly higher proportions [48].

the methodology used the predictive power of estimates was low and imprecise.[5] Therefore, it is probable that any t-step ahead forecasting exercise using past S3 data will be characterised by statistical insignificance and wide confidence intervals derived from noisy data.



(a) Percentage of FTSE All-World constituents disclosing Scope 1, 2, 3 emissions

(b) Percentage of 2021 reported S3 data within YoY variation thresholds

Figure 2.3: Global average temperature anomalies and greenhouse gas emissions, [25]

Lack of consistency and comparability

In addition to the issues already discussed, there are three reasons why S3 data may not be comparable across firms. First, despite the specifics covered in Section 2.3, the GHG Protocol provides firms with a wide room for manoeuvre regarding which accounting and reporting methodologies they use. For example, firms must make a number of choices including their organisational boundaries, which categories to report, the calculation approach for each category, data sources, how to allocate emissions, whether to report optional extras above the minimum boundaries and so on. As long as these choices can be justified according to the five key accounting principles (which are relevance, completeness, consistency, transparency, accuracy) then they are said to conform to S3 reporting standards [76]. This methodological latitude opens the possibility for meaningful heterogeneity in firms' S3 accounting and reporting: accordingly, while firms may be consistent with respect to themselves, and with respect to the GHG Protocol itself, aggregating firms together in a portfolio risks comparing apples and oranges.

Second, the GHG Protocol and S3 Guidance is often complimented with standard setters' frameworks and legal rules. While these may intend to be aligned to the GHG Protocol, they invariably introduce particular interpretations and requirements which their signatories are expected to adopt. This could lead to inconsistencies, especially between sectors as they may follow their own industry-specific practices. For example, many financial institutions have adopted the recommendations of the Partnership for Carbon Accounting Financials (PCAF) in an attempt to improve the reporting quality for Category 15 of S3 emissions [53]. However, as pointed out by

[5] See Figure 5.5 and in Section 5.1.4.

Tang et al. [66], the practical considerations of GHG estimation means that PCAF's framework does not align with the reality of availability, reliability and confidence of the "highest quality" corporate reported emissions data. This leads to increased non-standardisation, more inaccuracy and further concerns of inconsistency.

Third, when investors wish to compare the GHG emissions of many firms, perhaps to evaluate the transition risk of their portfolio, they frequently turn to data providers such as Bloomberg, CDP, ISS, MSCI, Sustainalytics, Thomson Reuters, and Trucost. However, due to missing data, these providers often use estimates to fill in the gaps. Unfortunately, each use their own methodologies with varying transparency. These different approaches can compound the inconsistency across the data and make it challenging for people to know what the "real" S3 emissions are for a collection of firms. Busch et al. [4] and Kalesnik et al. [38] found that third party estimated data was less consistent than data stemming directly from corporate reports, and so the choice of data provider might significantly affect empirical results.[6] Moreover, third party inconsistencies extend beyond differences in estimation methodologies. Even for values reported by firms, surprisingly Nguyen et al. [48] found only 68% of data points being identical between Bloomberg and Refinitiv Eikon.

2.5 Estimating S3 emissions

This Section reviews the academic and data providers' estimation strategies for S3 emissions. Unfortunately, there are not many quantitative papers on S3 emissions in a macro context, such as for a country or industry, due to the reasons mentioned in the previous Section [31].

As there is no industry consensus on the best approach for estimating GHG emissions [23], the literature offers several approaches, which Nguyen et al. [49] categorise as naive, empirical and ML. I employ this same framework to explore S3 estimation.

2.5.1 Naive estimation

First, there are naive approaches which use simple averages. They typically involve several sub-models which are then aggregated based on data availability. Common strategies include extrapolation from past emissions, using the firm's consumption or production figures, or simple sector/industry averages. While third party data providers often use more sophisticated models for S1 and S2, naive approaches are typically used for S3 estimation because they are easy to calculate, simple to explain, and sidestep some of the challenging data quality issues. Companies using these approaches include the ISS [36] and the London Stock Exchange Group (LSEG) [43].

[6] Busch et al. also found that this third party estimated data inconsistency was increasing over time, although this has been challenged with more recent data [65].

2.5.2 Empirical estimation

Second, empirical estimation comprises regression methods which model the linear relationships between variables. While this is a common approach in the literature for estimating S1 and S2 emissions, researchers often don't extend this approach to S3 due to data quality concerns [27, 38]. Nevertheless, a few attempts have been made.

The CDP uses a gamma generalised linear model using both revenue and activity information [8]. They choose the gamma distribution to accommodate the positive but heteroskedastic nature of the data. The target variables are average activity-group revenue intensities reported to the CDP, with revenue broken down into 44 variables. To prevent overfitting, model selection is optimised by pruning on activity groupings which are chosen to maximise the Akaike Information Criterion for each of the 15 S3 categories. Given the hundreds of possible variables that may be used, others have tackled model specification by using regularisation to automate feature inclusion [5].

A key drawback for these more traditional statistical approaches is that estimates rely on parametric assumptions that, if not met, will fail to provide the best, linear, and unbiased estimators. Especially important in the context of S3, Olesiewicz et al. [51] show that if the data is not missing at random, then the regression estimates will be biased.

2.5.3 Machine learning techniques

Thirdly, Nguyen et al. [48] explore the performance of ML models for predicting S3 emissions against naive and statistical approaches.[7] The two baseline models are an industry-fill model and a naive OLS model. The industry-fill model simply uses the median of disclosed emissions data of a firm's nearest available peer group. The naive OLS model predicts S3 emissions at a categorical level using just revenue, total employees, and a set of dummy industry indicators.

For linear models, given the limited number of S3 observations, using an extended set of predictors may lead to multicollinearity issues [49]. To avoid this, the authors use the forward-backward stepwise regression to automatically include the relevant predictors and exclude irrelevant ones. They further employ an elastic net to also address multicollinearity.

For their ML models, they use tree-based ensemble models due to the success of these kinds of models elsewhere in the emissions modelling literature [49, 29]. This success is in part because these models capture non-linearities and correlations well,

[7] At the time of writing this is the only previous ML paper on S3 estimation, and so this forms the main source of comparison with my own work. For the sake of completeness, there is also a paper by Serafeim and Velez Caicedo [60]. However, it remains a working paper only, has no citations, and reports extremely high predictive accuracy.

whilst also improving the stability and interpretability of the coefficients. Specifically, the authors employ both a Random Forest and Extreme Gradient Boosting. Finally, they also use a linear tree/forest approach which they hypothesise would lead to better predictive results because it combines the learning ability of decision trees with the extrapolation power of linear models.

The authors report big accuracy gains from predicting each category separately and then aggregating them up to estimate total S3 emissions for each firm. Upstream categories appear easier to predict than downstream categories. Surprisingly, out of all ML models, only the linear forest consistently outperforms the baseline models in predicting total S3 emissions using both aggregated data and category-level data. Furthermore, when considering the prediction accuracy of absolute emission values, even the best prediction model has substantial error. Specifically, the stepwise OLS generates a median absolute error of 72.3% on aggregated S3 emissions; this means that only 30% of the predicted emissions values are within +/- 50% of the actual reported emissions. Thus, the ML approaches tried so far appear disappointing both relative to simpler models and in absolute terms for predictive accuracy.

The Nguyen et al. paper [48], being the only past paper using ML techniques to estimate S3 emissions, will be the main point of comparison with respect to my own results in Chapter 5.

Chapter 3

Data

3.1 Source

This research project is in collaboration with the London Stock Exchange Group (LSEG). Accordingly, the data on which my analysis is based is sourced from LSEG Datastream, a global, historical, financial database covering all major asset classes and economic indicators. The data delivery mechanisms are a Desktop application, an Excel plug-in and a feed via an application programming interface (API) available through the REFINITIV-DATA Python package.[1]

I base my universe on the FTSE All-World Index; originally launched in the year 2000, as of June 2024 it covers 98 percent of the world's investable market by capitalisation and contains around 4300 large and mid-cap firms [24]. This index ensures a global coverage of firms across different industries. Moreover, I take a snapshot of constituents as of 2022-12-31 to retain a consistent set of firms in my sample.

3.2 Features and target variables

In line with previous empirical papers, I use firm fundamentals and characteristics to model and predict emissions. These features reflect aspects of a firm's size and nature of business, and these, in turn, relate to its carbon footprint.

There is often a tension between the availability of data and relevance to the model. For example, some have used the number of employees as a feature [49]; it is a useful proxy for a firm's size and it directly relates to some S3 categories such as employee commuting. However, it is not a standard metric in annual accounting and reporting standards, so the data is often unchanging for many years or missing altogether. Given the paucity of emissions data, I do not include features with over 10% of their values missing. This means that I do not include features such as: gross

[1]For further detail on LSEG Datastream see https://www.lseg.com/en/data-analytics/financial-data/company-data/company-fundamentals-data.

profit, current assets, current liabilities, inventories and cost of revenue, even though they may be theoretically relevant.

Table 3.1: Raw features queried from LSEG

Static features	Financials and fundamentals	Policy and emissions context	Emissions measures
Instrument	Market capitalisation	Policy emissions score	Total S1 emissions
Common name	Total revenue	Target emissions score	S1 estimation method
Country of risk country code	EBIT		Total S2 emissions
HQ country code	EBITDA		S2 estimation method
TRBC economic sector	Net cash flow		S3 purchased goods and services (cat 1)
TRBC business sector	Total assets		S3 capital goods (cat 2)
TRBC industry group	Receivables		S3 fuel and energy (cat 3)
TRBC industry	Net PPE		S3 transport and distribution upstream (cat 4)
TRBC activity	Capex		S3 waste generated (cat 5)
	Intangible assets		S3 business travel (cat 6)
	Long term debt		S3 employee commuting (cat 7)
			S3 leased assets (cat 8)
			S3 upstream estimation method
			S3 transport and distribution downstream (cat 9)
			S3 processing of sold products (cat 10)
			S3 use of sold products (cat 11)
			S3 end of product life treatment (cat 12)
			S3 leased assets (cat 13)
			S3 franchises (cat 14)
			S3 investments (cat 15)
			S3 downstream estimation method

Other features with insufficient coverage were removed as outlined in Section 3.3.

3.2.1 Feature descriptions

Feature construction

Table 3.1 lists the raw features I query from LSEG. This forms the basis for my sample construction. In a few cases, I perform simple aggregations to construct new features. First, I map the country of risk (or HQ country code if the former is missing) into either "Advanced" or "Emerging" buckets as set out by the International Monetary Fund (IMF) World Economic Outlook database for 2023 [34]. This is an important feature to include as, first, advanced economies have higher reporting standards and so higher rates of GHG disclosure, and second firms in advanced economies are likely to be less carbon intensive as their energy mix likely includes a higher renewable share and more modern technology. Some past studies have tried to use a firm's energy use, map that to their country's energy mix and include that in their feature set. However, energy use data is often missing and country energy mix data is infrequent. Therefore I have chosen to keep my model parsimonious by just using the country risk classifications of either "Advanced" or "Emerging", perhaps risking oversimplification and risking my models' prediction ability.

Second, I construct emission aggregates. I sum S1 and S2 emissions to capture a firm's GHG footprint outside of S3. I also combine the eight S3 upstream categories into a single S3 upstream value, and also do likewise for the seven S3 downstream categories. While Nguyen et al.'s study model each category, I have fewer data points in my sample [48]. In addition, category disclosure rates vary significantly with some disclosure rates very low (see Figure 3.2b). Therefore, while it would be ideal to model each category separately and aggregate up into S3 upstream and downstream,

this is not a viable approach in my case.[2] Therefore, I simply sum across the relevant categories. In turn, I sum the S3 upstream and S3 downstream totals into a single S3 category total.

Third, I calculate emission intensities by dividing the emission features by revenue. This means that an emission value, reported in metric tonnes of CO2 equivalent, can be transformed into an intensity, measured in metric tonnes of CO2 equivalent per million of USD revenue. In a world with full data availability, I would like to model the emissions profile of each component of a firm's activity. Lacking this, company emission intensity allows me to compare across firms more easily by normalising with size/activity.

Firm characteristics

A firm's carbon footprint is most likely to be tied to its scale of operations, therefore I use several measures of this. First, I use firm revenue, as carbon emissions are a by-product of revenue generation and this has been included in all prior models in the empirical literature. Second, I use the market capitalisation in the month closest to the firm's fiscal (i.e. reporting) period; market capitalisation can be thought of as an overall measure of the company's worth through investors' assessments of a firm's assets, liabilities and future cash flows. Third, the scale of operations in a firm may be indicated by its asset size. Therefore, I use total assets, intangibles, and net property, plant and equipment values. The latter is especially useful as high values may indicate the presence of physical plants and equipment (PPE) that are likely to be more closely related to GHGs. On the other hand, firms with high intangible assets are less likely to have a substantial carbon footprint.

Next, I include both earnings before interest and taxes (EBIT) and earnings before interest, taxes, depreciation and amortization (EBITDA). These can be useful proxies for business activity. For example, some firms with very stable cash flows are utility firms which may have a substantial carbon footprint. Meanwhile, other firms like commodity producers may exhibit highly volatile earnings depending on changes in spot prices. Having both EBIT and EBITDA also allows me to consider the impact of asset depreciation; higher depreciation might mean firms invest in more equipment, thereby modernising, and so reducing their carbon footprint. I also include other earnings metrics such as net cash flow and receivables.

In addition, I include a firm's long term debt, as I suspect that higher long term debt restricts a firm's ability to borrow, invest, and modernise, which may raise their carbon footprint. Most importantly for the business model, I include a dummy for the firm's peer group, which is an indicator of a firm's economic sector, industry, or activity depending on the appropriate level of granularity. This is described in more detail in Section 3.5.

[2]Nevertheless, in Section 6.3 I discuss category-level modelling further and attempt to model category one.

Capital expenditures (capex) is another important variable I include. Adoption of modern technologies and equipment can be indicated by high levels of capex. As such, it is a feature that goes some way to explain a firm's level of technology which has clear implications for its carbon footprint via efficiencies.

Environmental and regulatory context

The final features I include concern the firm's environmental and regulatory context. I begin with each firm's so-called primary country of risk. According to Refinitiv, this uses the StarMine Countries of Risk Model which is based on four sources of data: its revenue distribution by geography, the location of a company's headquarters, the country where its primary equity security listing trades, and its financial reporting currency. Together, this gives a rounded view of which country the company is most exposed to.[3] I then bucket this variable into "advanced" or "emerging" economy dummies as aforementioned.

I also include measures relating to the firm's policies on emissions reduction which situates the firm's efforts within its sector. The Target Emissions Score measures to what extent firms have set targets on emissions reduction. The Policy Emissions Score measures how the firm uses its policies to improve emissions reductions; this includes measures relating to the mechanisms or programmes in place, using a firm's formal, documented processes for controlling emissions and driving improvement. Both of these emission scores range from 0 to 100 and follow a transparent, detailed, fully-automated methodology [28].

As well as the features outlined above, there are production-specific features that I considered but, on reflection, decided to exclude. These include measures such as how much crude oil or natural gas is produced. Such measures only apply to a very small number of firms in the sample, and while this would provide more colour on these firms, it clearly would not extend to building a general model that spans across sectors.

3.2.2 Target variable

I model three target variables for S3 emissions: total, upstream and downstream (refer to Section 3.2.1). Lastly, my final target variable is total S1 and S2 emissions. This allows me to compare against findings from earlier studies which only model emissions for S1 and S2. It also provides me a way to see if there are any major differences between S1 and S2, and S3 modelling. While S1 and S2 emissions are conceptually distinct from S3, a simple plot of total S1 and S2 vs total S3 emissions reveals, surprisingly, a significant, positive relationship (see Figure A.1 in Appendix A). Following this relationship, I found that including S1 and S2 emissions improves

[3]This is a better measure than the HQ location which may be skewed by tax regimes.

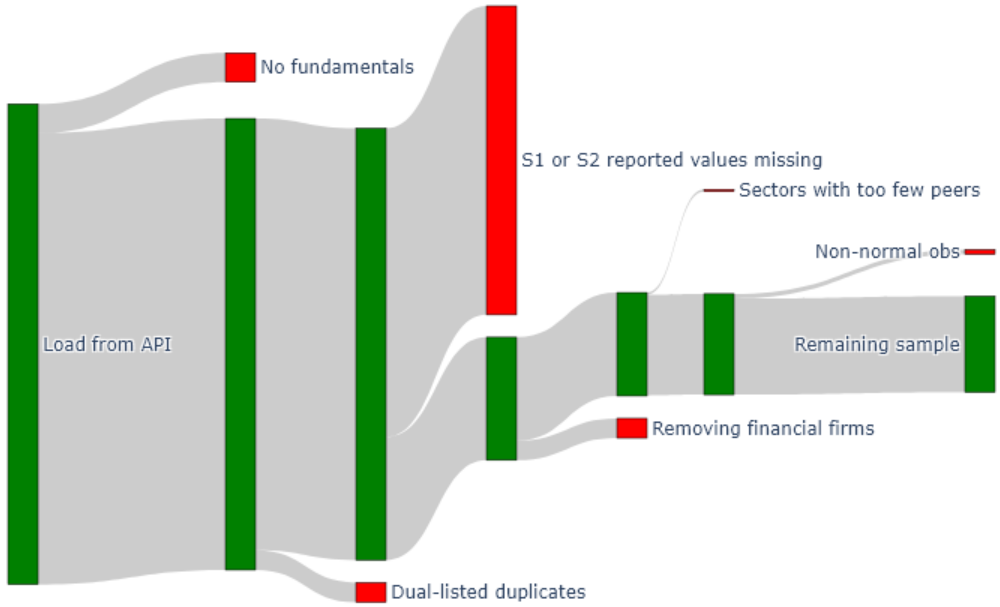


Figure 3.1: Sankey diagram of sample construction

predictive accuracy for S3 emissions. Therefore, for all S3 target variables, S1+S2 emissions are also included as a feature.

3.3 Sample construction

Ahead of modelling GHG emissions, I do some work to prepare my data. First, I call the data from the Refinitiv database using the API wrapped in the Python package REFINITIV. To be assured of the results, I also separately call the data via the Excel plug-in and perform a systematic comparison to ensure they return identical results.

I begin by retrieving the static data on all FTSE All-World index constituents as of 2022-12-31, such as primary country of risk and sector information. I then merge on a set of fundamentals data. Finally, I add on a snapshot of each firm's market capitalisation using the nearest available (backward-looking) data point relative to the fundamental data.

With the raw data loaded, I undertake a series of filtering processes, visualised in the Sankey diagram in Figure 3.1. There are three important aspects of this process. First, most of the filtering occurs by removing observations without S1 and S2 data. This shows that even with the better covered scopes, coverage isn't guaranteed as reporting stringency varies according to region. Second, I have chosen to remove all financial firms as their fundamentals are often not an accurate reflection of their underlying business, and it is common practice to consider only non-financial firms in cross-sector analysis. Third, as a final stage of filtering I remove observations

with abnormal revenue or CO2e values based on year-over-year ratios. Specifically, I identify instruments with extreme changes in revenue, S1 and S2 CO2e, S3 upstream CO2e, and S3 downstream CO2e by calculating the ratio of maximum to minimum values over time for each instrument. Then, observations with ratios exceeding the 99th percentile for any of these metrics are considered abnormal and removed from consideration.

In addition, I took steps to improve data coverage and numerical stability of some of the features. Fundamentals data were forward and backward filled to account for missing values. In addition, net cash flows were winsorized given the high skew and kurtosis exhibited.

3.4 Coverage

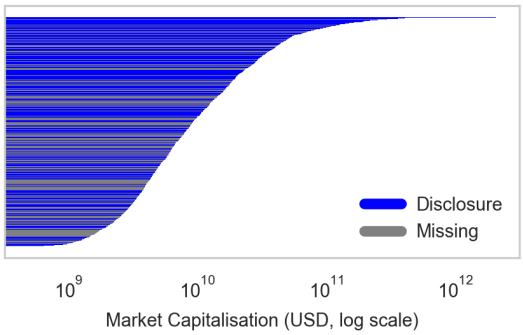
One of the main concerns discussed in Section 2.4 is data coverage. As shown by the summary of the numeric variables given in Appendix C, while the full sample (taking just S1 and S2 data points) has around 10,800 observations, S3 category-level data only exists for around 3,200 observations. To understand this in more detail, Figure 3.2 comprises four subplots to illustrate how S3 emissions are covered in my sample.

Figure 3.2a uses a snapshot of the full sample in 2022, showing how S3 disclosure improves with firm market capitalisation. This makes intuitive sense as larger firms likely have more resources (and arguably more investor pressure/profile) to disclose their S3 data. However, it also corroborates the argument that carbon emissions data is likely missing not at random [51].

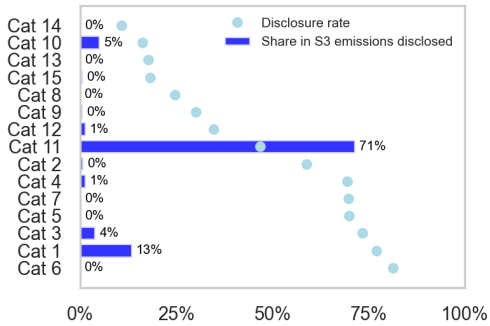
Second, Figure 3.2b considers all the S3 category-level data reported in my sample. It reveals how firms tend to disclose what is easy to disclose, rather than what is material to their emissions profile. For example, category 6 of S3 emissions refers to emissions from employee business travel. While this category has a disclosure rate of over 80 percent, it accounts for less than one percent of total S3 emissions reported.[4] The most material categories with respect to how they impact total S3 emissions are category 11 (use of sold products) with 71 percent, category 1 (purchased goods and services) with 13 percent, category 10 (processing of sold products) with 5 percent, and category 3 (fuel- and energy-related activities) with 4 percent.

Ideally, S3 category disclosure would better reflect how material each category is for building an accurate picture of a firm's emission profile (see the discussion in Section 2.4.1). To check if this be the case, for each economic sector I map which two categories are most material as recommended by the S3 FTSE Russell Report [25]. Accordingly, my sample materiality is shown in Figures 3.2c and 3.2d.

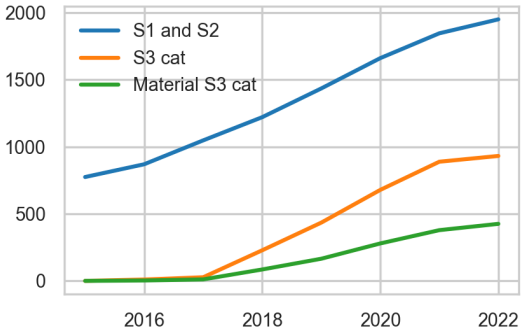
[4]Disclosure in this context refers to: for a given firm-year observation, if S3 categories are disclosed, what percentage of them include a given category.



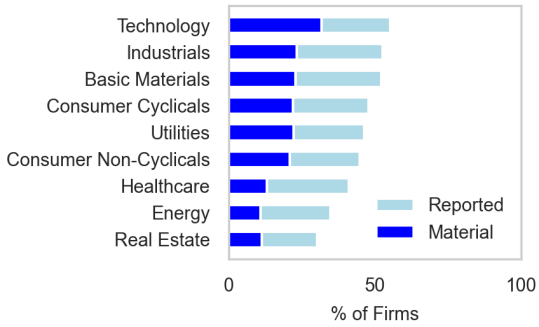
(a) S3 disclosure by market capitalisation



(b) S3 disclosure by category



(c) GHG emissions reporting by year



(d) S3 disclosure by sector

Figure 3.2: S3 coverage in my sample, inspired by [23] and [25]

Figure 3.2c shows a simple count of observations per year. As the instruments I use for my universe are the 2022 constituents of the FTSE All-World Index, the number of instruments gradually reduces the further back in time as some firms had not yet listed. Nevertheless, a clear pattern emerges which is consistent with Figure 2.3a. First, the S3 emissions are consistently lower than S1 and S2 emissions. Second, material S3 disclosure account for only around half of total S3 disclosure. Third, it is clear that pre-2020, S3 category reporting is somewhat rare. These observations inform my modelling approach set out in Section 4.4.

Finally, Figure 3.2d reports what percentage of firms disclose S3 category data and the split between total disclosure vs material disclosure by economic sector. While technology firms disclose most and have a majority of their disclosures material, the healthcare, energy and real estate sectors all exhibit relatively low rates of disclosure and low shares of material disclosure. This clear imbalance between sectors raises the question of how to deal with firms from different industries, which leads me to consider my stratification strategy.

3.5 Sample stratification into peer groups

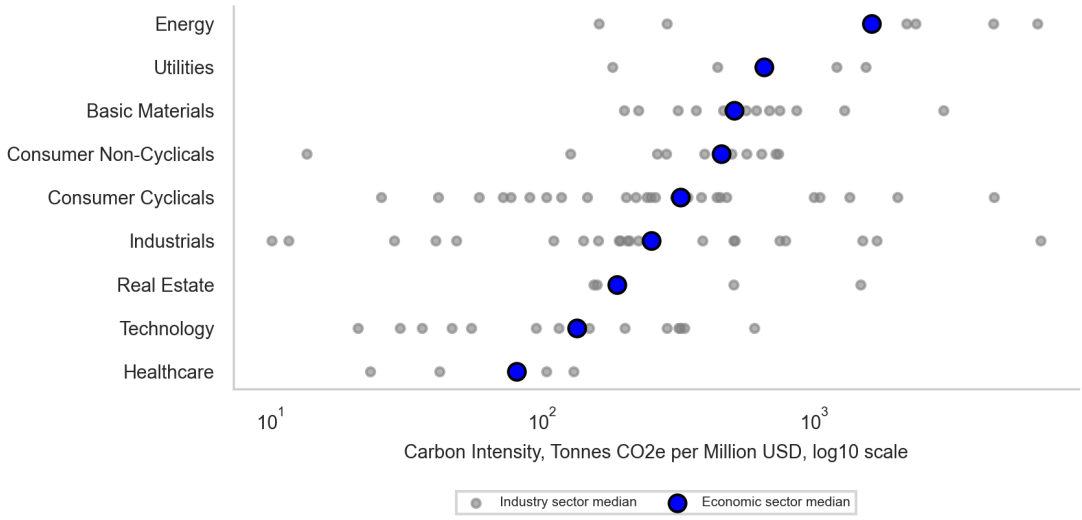
Firms belong to different sectors which influences their GHG emissions as discussed in Section 3.3. Effective stratification ensures that both the training and test sets maintain similar proportions of firms from each sector. Without this stratification, a random split could result in an under- or over-representation of certain sectors in the test set, potentially compromising model performance. By stratifying the data to mirror the real-world distribution, I can enhance the model's ability to generalise, especially when sector classes are imbalanced (see Figure 3.3b).

The Thomson Reuters Business Classification (TRBC) has five levels of sector with increasing granularity: economic sector, business sector, industry group, industry, and activity.[5] On the one hand, using more granular levels can potentially improve the accuracy of predictions; Figure 3.3a illustrates that even within an economic sector there is a large variation in constituent industries. While variation generally decreases with TRBC level, Figure 3.3c shows how it comes at the cost of reducing peer group size.

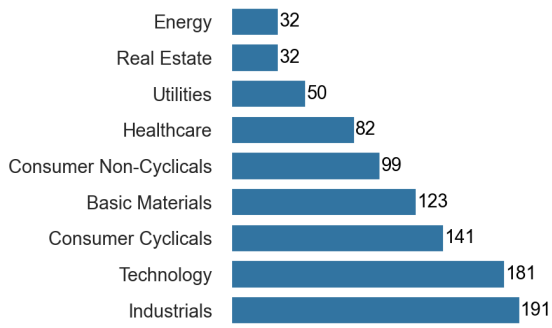
To select an appropriate peer group, I use an algorithm which takes as inputs the number of folds to create along with the minimum number of peers per fold. The algorithm then determines the most granular TRBC classification which I assign to the firm's "peer group". For example, if I require at least three firms per fold, and ten folds, then the algorithm finds the lowest level sector that has at least thirty firms.

With this preparatory work complete, the modelling can follow.

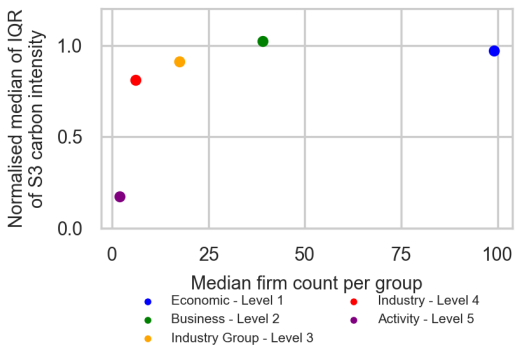
[5] See here for more details.



(a) S3 emission intensity by economic sector and industry subgroup



(b) Economic sector breakdown



(c) Sector coverage vs volatility

Figure 3.3: Sector plots for firms reporting S3 emissions in 2022

Chapter 4

Modelling approach

Chapters 2 and 3 have set out the policy background, previous empirical efforts, and some high-level data exploration. This Chapter introduces the four key dimensions that will determine the specifics of my modelling approach. As a simple framework to orient the discussion, consider the regression problem in abstract in Equation 4.1:

$$\mathbf{y}_{it} = f\left(\mathbf{X}_{it}\right) \tag{4.1}$$

My goal is to predict S3 emissions using an appropriate **target variable** $(\mathbf{y})$, a **set of explanatory features** $(\mathbf{X})$, over a **set of observations** (i, t), using the best **model** f. I consider each of these dimensions in turn.

4.1 The target variable

There are four targets to predict: S3 upstream emissions, S3 downstream emissions, total S3 category emissions, and (for comparison), total S1 and S2 emissions. There are two ways to measure these targets. First, I could use absolute emission levels. Second, I could use emission intensity (absolute emissions divided by revenue).

As a first step in my empirical approach, I will try modelling with absolute levels versus intensities and compare the results in predictive ability.

4.2 Rationalising the feature choice set

The second important choice is which set of explanatory features to use. The choice of features is driven by a combination of both data availability and *a priori* theory. One potential issue is the presence of multicollinearity i.e. where several features are highly correlated with each other. Indeed, the correlation matrix presented in Figure 4.1 reveals that some variables are highly correlated, e.g. EBIT and EBITDA. This has the potential to cause computational problems and makes precise estimation of coefficients more difficult.

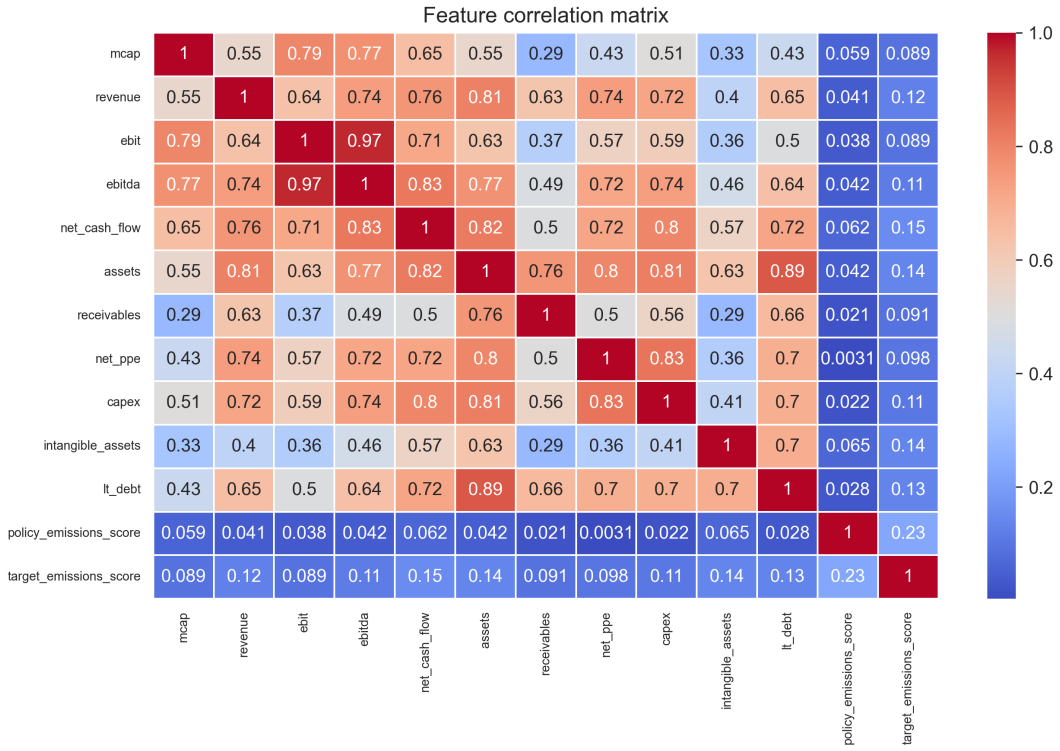


Figure 4.1: Correlation matrix between numeric features

An attempt to deal with this issue from Nguyen et al's previous study uses a Forward-Backward Stepwise Regression, which automatically includes relevant predictors (e.g. $< 1\%$ significant level) into the model and excludes irrelevant ones (e.g. $> 5\%$ significant level) [48]. When I take this algorithmic approach to feature selection I retain only 15 features, with the vast majority of dummy variables removed. While on the surface this may appear to be an attractive solution, as pointed out by Harrell (2015) this "violates every principle of statistical estimation and hypothesis testing" [30].

Another common tactic to select relevant features is to use feature attribution (FA) methods. These methods are popular in explainable AI to help people understand how the inputs of a ML model contribute to its output. Common methods include perturbation-based ones such as `LIME` [56] and `SHAP` [44], or gradient-based ones such as Integrated Gradients [64] and `DeepLIFT` [61]. These approaches reduce the complexity of a model's mathematical logic into a set of numerical importance scores for each feature.

Unfortunately, these FA methods cannot be straightforwardly applied to time series data. This is because there are temporal dependencies in time series data meaning that data points are dependent on previous values. By contrast, methods such as LIME or SHAP are designed for independent and identically distributed (i.i.d.) data. By permuting or masking certain features, they break this dependency and produce results that are unrepresentative of true behaviour.

Moreover, by evaluating individual features, these FA methods often overlook how the sequence of past observations contributes to the current value, in effect ignoring the data's autocorrelation. Furthermore, in a time series context, feature importance may vary over time, so the feature attribution becomes dynamic rather than static.

Given that stepwise regression and standard FA methods are not statistically valid approaches to feature selection, I opt for a different way. From the set of all possible features described in Section 3.2.1 with sufficient coverage, I create three sets: the extended set, the core set, and the limited set, as described by Table 4.1.

Table 4.1: Feature sets used in the analysis

Extended feature set	**Core feature set**	**Limited feature set**
Market cap, revenue, EBIT, EBITDA, net cash flow, assets, receivables, net PPE, capex, intangible assets, long term debt, policy emissions score, target emissions score, scope 1 and scope 2 total emissions	Market cap, revenue, capex, intangible assets, long term debt, scope 1 and scope 2 total emissions	Market cap, revenue, scope 1 and scope 2 total emissions
Year dummies, Peer group dummies, Country classification dummy		

The extended set of features includes the whole range of metrics in Section 3.2.1. The core set includes six key measures, and the limited set only three. All sets include S1 and S2 emissions as well as a group of dummy variables. Similar to selecting the target variable, for selecting which features to use I try all different combinations to see which sets, in general, produce results with higher predictive accuracy.

As well as feature choice, feature scaling is important in many ML algorithms. This is because they often use distance-based measures (e.g. K-Nearest Neighbours), gradient-based optimisations (e.g. neural networks) or penalisation methods (e.g. elastic nets); without scaling these models may fail to find the optimal solution or may take a very long time to converge.

Log-scaling is often used in financial settings due to the skewed nature of the data; this maps orders of magnitude differences into a linear scale. I use log-scaling for the core feature set and the limited feature set. For the extended feature set, I instead use robust scaling (value minus the median, divided by the interquartile range) as some of the cash flow variables such as EBIT may be negative, and the log function is not defined for that domain.

4.3 Observation subsets

In normal circumstances, I would use all data to train or test my models. However, one worry is that the impact of the Covid-19 pandemic may feed anomalous and unrepresentative observations into the model training, reducing predictive ability.

Another major concern about S3 data is that firms may only partially report, leaving open the potential that they provide S3 figures that do not accurately capture their true carbon footprint.

To mitigate these concerns, I use three different observation sets. My first set simply includes all observations over the date range (2016 to 2022). My second set ("Ex-Covid") excludes all observations from the fiscal year 2021. This is an admittedly crude approach because different firms and industries were impacted by Covid at different times. However, lacking the data to incorporate this more precisely, and given the non-standardised reporting dates of firms' disclosures, this provides me an acceptable approximation.

Finally, to tackle the problem of under-reporting, LSEG has proposed "material" S3 categories for firms in each sector [25].[1] This picks out the two most important S3 categories for each firm peer group. For example, they report that for technology firms, 88 percent of their S3 intensity is composed of their two material categories. Therefore, as a third set of observations to experiment with, I use the subset that contain non-missing values in material categories for the firm's peer group. This also has the desirable property of trying to predict material emissions, and not merely what firms report as their emissions. Hence, such predictions could be used as a basis to guard against under-reporting in line with a precautionary principle (see Section 6.2 for a more in-depth discussion on this point).

To calculate which sectors are material, I follow a similar approach to LSEG in their report [25]. First, I calculate the median S3 intensity for each S3 category in each TRBC business sector. Next, I rank these medians and designate the two largest the "material" S3 categories for that business sector. Finally, for each firm, I flag that they are material reporters if they report the two highest S3 categories for their business sector. To mitigate against underestimating material categories with low reporting rates (see Figure 3.2b), this is performed on all available reporting companies from each category. For most business sectors, the material categories are purchased goods and services, and use of sold products (that is, S3 categories 1 and 11 respectively. See B.1 for a full breakdown.)

4.4 Model choice

The final key choice concerns which model type to use. In this Section, I start with introducing my end-to-end process for training and testing all the models. I then move on to consider each of the models in turn.

[1] See also [9] for a technical note on the relevance of S3 categories by sector.

4.4.1 Modelling workflow

My empirical process is visualised in Figure 4.2. I begin by loading and cleaning the data to construct my sample, as described in Section 3.3. In addition, I undertake some important preprocessing steps such as feature scaling and aggregation.

Next, I stratify my sample (see Section 3.5) into ten folds with at least three unique firms per peer group in each fold.[2] This produces a roughly balanced number of firm-year observations for each peer group. One fold is pseudo-randomly chosen as the test dataset and set aside until the final evaluation. The remaining training data is passed into a k-fold cross-validation over a grid-search of hyperparameters.[3] Once the best combination of hyperparameters has been found, the model is then re-trained using all the training data and the chosen hyperparameters. Finally, this trained model is applied to the held-out test fold. Most of the trained models are implemented using the Python package SCIKIT-LEARN.

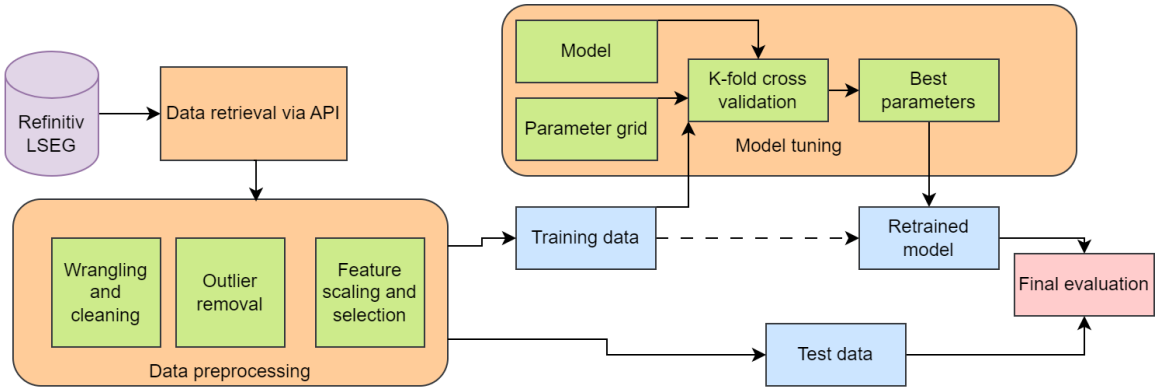


Figure 4.2: Empirical workflow

Note: my code base, summarised by this diagram, can be found at https://github.com/Michael-Hollins/Imperial_MSc_AI/tree/main/CarbonForecaster

4.4.2 The models

Having covered the general modelling process, it remains to introduce the eight model types I employ to predict GHG emissions. In keeping with the labels of "naive", "statistical" and "machine learning" used from earlier literature (see Section 2.5), I use two naive models, two statistical models, and four ML models. I anticipate that eight model types may, at first, appear indiscriminate. However, each brings its own unique set of strengths and weaknesses, such that each helps add to the overall picture of what works better or worse. In addition, for regression problems such as this, many techniques have been developed by the ML community precisely because there is not a one-method-fits-all paradigm. Furthermore, estimating GHG emissions

[2]For selecting material observations only, I use seven folds with at least two unique firms per fold due to the lower number of observations.

[3]Details of the key hyperparameters over which cross-validation was performed - and which were subsequently chosen following model tuning - can be found in Appendix D.

is still a nascent field, and I believe it is reasonable to try many different approaches to see what performs better.

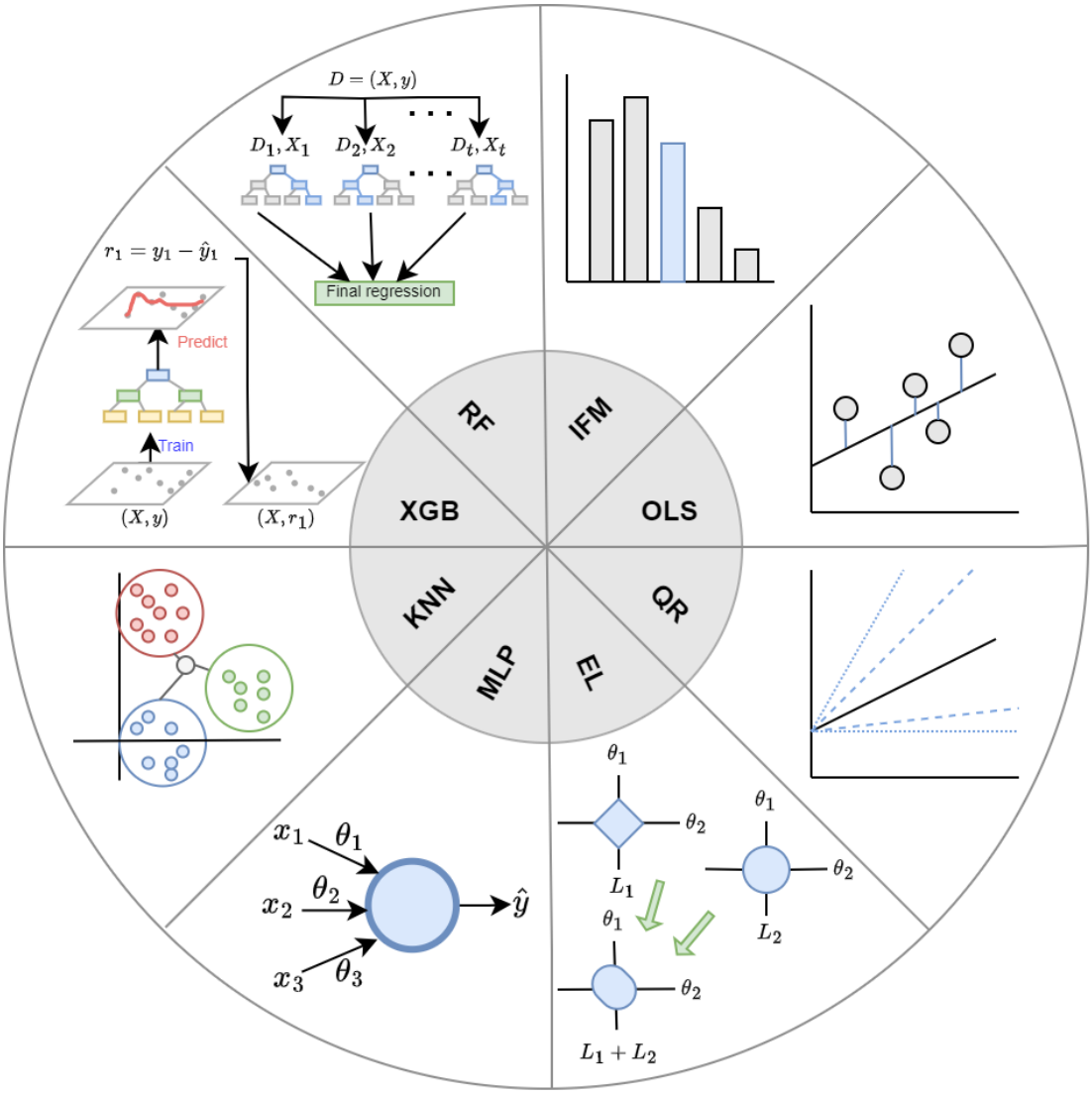


Figure 4.3: All models used to predict S3 emissions

Key: IFM = industry-fill model, OLS = ordinary least squares, QR = quantile regression, EL = elastic net, MLP = multi-layer perceptron, KNN = k-nearest neighbour, XGB = extreme gradient boost, RF = random forest

Moreover, it is also important to consider, briefly, the model types I do not use, and how this relates to my overall approach to time series forecasting in this context. There are two common approaches to time series forecasting. First, there is "direct forecasting" which involves predicting future values of a time series by using past observations. In other words, the goal is to estimate the value of the target variable at time $t + i$ based on the data available up to time t. Second, there is forecasting with covariates whereby a model learns the relationship between the target variable

and one or more covariates during a training phase. In the test phase, the model predicts the value of the target variable based on the values of the covariates. This is also sometimes termed "multivariate time series forecasting".

It is clear from Section 3.4 that, because coverage has greatly improved in later years, the approach of direct forecasting would risk reducing the training dataset too much because I would lose a lot of observations by even excluding just the year 2022. Due to the nature of the data available, therefore, I have adopted the second approach of forecasting with covariates. In order to use methods that respect the characteristics of my data, I have also excluded certain models from consideration.

Firstly, I have avoided autoregressive integrated moving average (ARIMA) and vector autoregression (VAR) models which are common time series forecasting approaches in other domains. This is because these models are better at handling small, less noisy datasets. Additionally, they were developed to model univariate processes or only a handful of variables. Secondly, I have avoided using transformer-based deep learning models, such as the temporal fusion transformer (TFT). This is because these models are typically better at capturing more intricate time dependencies such as trend, seasonality and ad-hoc components such as holiday periods. By contrast, my emissions data is annual, limited, and too volatile to exhibit any clear trend. In addition, the more established deep learning approaches such as RNNs and LSTMs were designed with direct forecasting in mind which also makes them unsuitable for application in this context.

4.4.3 Naive models

I use two naive models - the industry fill and OLS regression - which provide simple baselines against which to benchmark the more sophisticated models to follow.

Industry fill

Beginning with the most simple, the industry-fill model (IFM) estimates the carbon emissions of a firm by using the median among its peer group in the training data.[4] This provides a baseline against which to assess more sophisticated techniques, and it is a typical benchmark used in other studies such as in Klaaßen and Stoll [39] and Nguyen et al [49, 48]. Just as importantly, it is also a standard method used by the major commercial data providers such as LSEG [25] and Bloomberg [2], at least as a starting point.

The IFM for total metric tonnes of CO2 equivalent for firm i in peer group IND at time t is:

[4]The name "industry-fill" is found elsewhere in the literature because it is typically the firm's industry from which the median is derived. In my context, I also use the firm's nearest available peer group as described in Section 3.5. However, to remain consistent with the literature I use the term "industry-fill" rather than "peer-fill".

$$Emission_{i \in IND,t} = median(Emission_{IND}) \tag{4.2}$$

The IFM has several advantages: it is tractable, easy to compute, the data is widely available, it is not skewed by outliers, and it aligns with the 'precautionary principle' emphasised in the literature [32, 50], such that firms which don't report are assumed to be at least as emitting as their peers who do report.

However, there are plainly limitations to the IFM. First, it ignores all firm-specific information which may be relevant, such as firm size. Second, even within peer groups there is still a wide range of GHG emission values. Notwithstanding these drawbacks, it is a useful starting point.

OLS Regression

Ordinary least-squares (OLS) regression is a technique that estimates the conditional mean of the dependent variable given some independent variables. Put another way, it models the average relationship between the predictors and an outcome using a single line. This line represents the best fit for all data points based on the mean of the dependent variable. OLS is one way to derive the line of best fit.

Another common baseline is OLS regression based on a proxy for firm size and some fixed effects. For example, Kalesnik et al. [38] regress log emissions on log of net sales along with industry, year and country fixed effects, finding that this explains over 80 percent of in-sample variation.

Formally, let y denote the dependent variable represented as an $n \times 1$ vector of observed values, X an $n \times m$ matrix of explanatory variables, and β an $m \times 1$ vector of coefficients to be estimated. OLS fits a parameter vector β such that the predicted values $\hat{y} = X\beta$ are as close as possible to the observed values in y. This is done by minimising the sum of the squared residuals (i.e. differences) between the observed values y and the predicted values $\hat{y}$. Accordingly, the goal of OLS is to minimise the objective (loss) function:

$$\text{minimise } \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \text{minimise } (y - X\beta)^\top (y - X\beta), \tag{4.3}$$

To find the vector β that minimises the sum of squared residuals, one takes the derivative of the objective function with respect to β and sets it to zero:

$$\frac{\partial}{\partial \beta}\left((y - X\beta)^\top (y - X\beta)\right) = -2X^\top (y - X\beta) = 0. \tag{4.4}$$

Solving for β, one gets:

$$X^\top X\beta = X^\top y \Leftrightarrow \beta = (X^\top X)^{-1} X^\top y \tag{4.5}$$

Assuming that the matrix X is of full-rank and therefore invertible.[5]

As it is intended to provide a second, simple baseline, I use the following model for estimating S3 emissions by OLS:

$$\begin{aligned} logEmission_{i,t} = \beta_0 + \beta_1 logRV_{i,t} + \beta_2 logMCAP_{i,t} + \beta_3 log(S1 + S2)_{i,t} \\ + \sum_j \gamma_j IND_j + \sum_t \delta_t YEAR_t + \theta CC + \epsilon_{i,t} \end{aligned} \tag{4.6}$$

In this equation, $logEmission_{i,t}$ represents the log of S3 emissions for firm i in year t, which is the dependent variable. β_0 is the intercept, representing the baseline level of $logEmission_{i,t}$ when all other variables are zero. $logRV_{i,t}$ is the log of the revenue for firm i in year t, and β_1 is its associated coefficient, capturing the effect of revenue on emissions. $logMCAP_{i,t}$ is the log of the market capitalisation of firm i in year t, with β_2 representing the effect of firm size on S3 emissions. $log(S1 + S2)_{i,t}$ represents the log of combined S1 and S2 emissions, and β_3 captures the contribution of direct and energy-related emissions to the prediction of S3 emissions. $\sum_j \gamma_j IND_j$ includes a set of industry dummy variables (IND_j) for firm i, where γ_j are the coefficients that control for industry-specific effects. $\sum_t \delta_t YEAR_t$ represents year fixed effects, with δ_t as the coefficients controlling for temporal variations. CC is a dummy variable that indicates whether a firm belongs to an advanced or emerging economy, with θ as its coefficient, capturing the effect of country classification on emissions. $\epsilon_{i,t}$ is the error term, accounting for the variability in emissions that is not explained by the model's predictors.

OLS will act as a second baseline. However, given the objective function is based on the ℓ_2 norm it is likely to be undermined by the presence of outliers. This leads me to consider the statistical models which employ regularisations.

4.4.4 Statistical models

Quantile Regression

Whereas OLS regression estimates the conditional mean of the dependent variable given the independent variables, quantile regression (QR) estimates the conditional quantiles of the dependent variable given the independent variables. Rather than focusing on just the central mean tendency, QR allows one to model the distribution of the independent variable. This is useful in the case when the data distribution is highly skewed by outliers and so the mean can be distorted by extreme values. It is also helpful in cases where the effect of predictors is not uniform in the distribution.[6]

Unlike OLS which minimises the sum of the squared residuals, QR minimises the

[5] In practice, inverting the X matrix can be computationally expensive, so SCIKIT-LEARN does the optimisation algorithmically instead.

[6] Also known as "heteroscedasticity", where the variance of residuals changes across the magnitude of the independent variable.

sum of the weighted absolute residuals. Therefore, for the τ-th quantile $(0 < \tau < 1)$ the QR model tries to find the coefficient vector β such that: $\hat{y}_i = X_i\beta$. This minimises the objective function:

$$\min_{\beta} \sum_{i=1}^{n} \rho_\tau(y_i - X_i\beta) \tag{4.7}$$

where $\rho_\tau(u)$ is the check-function defined as:

$$\rho_\tau(u) = \begin{cases} \tau u & \text{if } u \geq 0 \\ (\tau - 1)u & \text{if } u < 0 \end{cases} \tag{4.8}$$

As QR minimises the ℓ_1 norm, all errors are treated with equal magnitude, unlike the ℓ_2 norm in OLS.

Elastic Net Regression

The elastic net (EL) model blends both ℓ_1 and ℓ_2 norms for regularisation [78]. The ℓ_1 norm, taken from the Least Absolute Shrinkage and Selection Operator (LASSO), adds a penalty term to the loss function proportional to the absolute value of the coefficients. This penalty term encourages model sparsity, driving some coefficients to zero; this effectively performs variable selection by excluding irrelevant variables.

The presence of the ℓ_2 norm regularisation, as in Ridge regression, discourages large coefficients and tends to shrink them towards zero. However, unlike LASSO, it doesn't make them exactly zero and so all features remain in the model.

There are two important parameters in the EL. First, α controls the strength of the regularisation; setting $\alpha = 0$ is equivalent to solving with OLS. Second, λ acts as the mixing parameter and controls to what extent the EL is a mixture of LASSO and Ridge. Accordingly, the optimal β is found through minimising the loss function:

$$\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - X_i\beta)^2 + \alpha \left(\lambda \|\beta\|_1 + \frac{1-\lambda}{2} \|\beta\|_2^2 \right) \right\} \tag{4.9}$$

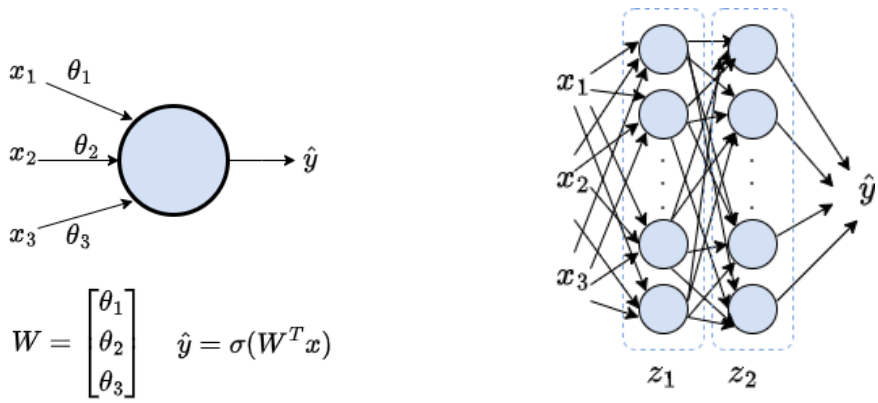
The advantages of the EL model are that, first, it is flexible because adjusting the λ parameter can make the transition between Ridge and LASSO easy according to the needs of each dataset. Second, it tends to select groups of correlated variables together which helps mitigate instability. Third, it is particularly useful when dealing with highly correlated features as flagged earlier in Section 4.2.

4.4.5 Machine learning models

So far, all the models considered have been linear. The use of ML models allows me to capture non-linearities in the data. In theory, this might improve predictive power, but previous studies have shown that the gains using the latest ML models may be modest [49, 48].

Multi-layer Perceptron

As the name suggests, the multi-layer perceptron (MLP) model is a type of artificial neural network that is composed of perceptrons stacked into layers. The leftmost layer, known as the input layer, consists of a set of neurons representing the input features. The rightmost layer is the output layer. The layers in between are known as hidden layers which take inputs from the previous layer, process them, and pass them on to the next layer.



(a) Artificial neuron

(b) A two-layer fully-connected MLP

Figure 4.4: Stylised depiction of MLP structure

Source: Adapted from Imperial College London's Introduction to Machine Learning, Lecture 4 notes [14]

Put more formally, MLP is a supervised learning algorithm that learns a function $f : R^m \to R^n$ by training on a dataset where m is the number of features and n is the dimension of the output. Therefore, given the target vector y and the feature vector $x = x_1, x_2, \ldots, x_m$, it can learn a non-linear approximator for regression. The non-linearities are introduced through the activation functions. Using a non-linear activation function σ, a two-layer MLP is given by:

$$\begin{aligned} \mathbf{z}^{(1)} &= \sigma(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \\ \mathbf{z}^{(2)} &= \sigma(\mathbf{W}^{(2)}\mathbf{z}^{(1)} + \mathbf{b}^{(2)}) \\ \mathbf{y} &= \mathbf{W}^{(3)}\mathbf{z}^{(2)} + \mathbf{b}^{(3)} \end{aligned} \tag{4.10}$$

where $\mathbf{W}^{(1)}$ is the weight matrix connecting the input layer to the first hidden layer and $\mathbf{z}^{(1)}$ is the output of the first layer etc. For regression applications there is no activation in the output layer.

The weights and biases are learnt through backpropagation i.e. gradient descent applied to a computational graph. It iteratively applies the chain rule of calculus to propagate the error backward through the network layer by layer. For a loss function L and a learning rate η, the weights are updated via:

$$\mathbf{W}^{(l)} \leftarrow \mathbf{W}^{(l)} - \eta \frac{\partial L}{\partial \mathbf{W}^{(l)}} \tag{4.11}$$

Through the use of hidden layers, the MLP can learn different layers of abstraction in the data through complex non-linear relationships. This is a key advantage over the naive and statistical models.

K Nearest-Neighbours

K Nearest-Neighbours (KNN) is a non-parametric approach that makes predictions based on the closest available data points in the feature space [62]. Using a distance measure, it finds a data point's closest neighbours in the training data and returns an average value for the target variable. As KNN doesn't assume any functional form between the features and the target, instead using only observation similarity, KNN inherently captures non-linear relationships. It has other advantages too: it is easy to implement, understand, and interpret.

Unfortunately, KNN exhibits the *curse of dimensionality*: as KNN relies on distance metrics, it may struggle when modelling many features in high dimensional spaces. Specifically, if many features are irrelevant, instances that should be near to each other may instead be far away. Notwithstanding this problem, it remains a useful alternative regression approach due to its non-parametric nature.

Regarding hyperparameters, the performance of KNN heavily depends on the choice of k, i.e. how many neighbours are taken into account. I optimise the choice using cross-validation, testing k values from 5 to 500 with intervals of 5.

Another key choice is which distance measure to use. I try three measures: for two points $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$: the Euclidian distance is given by $d_{\text{Euclidean}}(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$. The Manhattan distance is given by $d_{\text{Manhattan}}(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} |x_i - y_i|$. Finally, the Minkowski distance with parameter p is $d_{\text{Minkowski}}(\mathbf{x}, \mathbf{y}) = (\sum_{i=1}^{n} |x_i - y_i|^p)^{\frac{1}{p}}$.

XGBoost

In the only two previous studies that have employed ML techniques to estimate GHG emissions, tree-based methods appear to have shown the most promise [49, 48]. Moreover, decision trees are scale invariant, robust to irrelevant features and are readily interpretable. Therefore, the last two models I include in my set are both tree-based: XGBoost and Random Forest.

Gradient boosting is an iterative ensemble method where the models are built in sequence, each one correcting the errors of its predecessor. It does this by fitting a decision tree to the residuals (errors) of the previous model and then adding this tree to the ensemble. Through this process, gradient boosting gradually builds a strong predictive model by combining the outputs of multiple decision trees. This effectively captures complex interactions between features and handles non-linear relationships in the data.

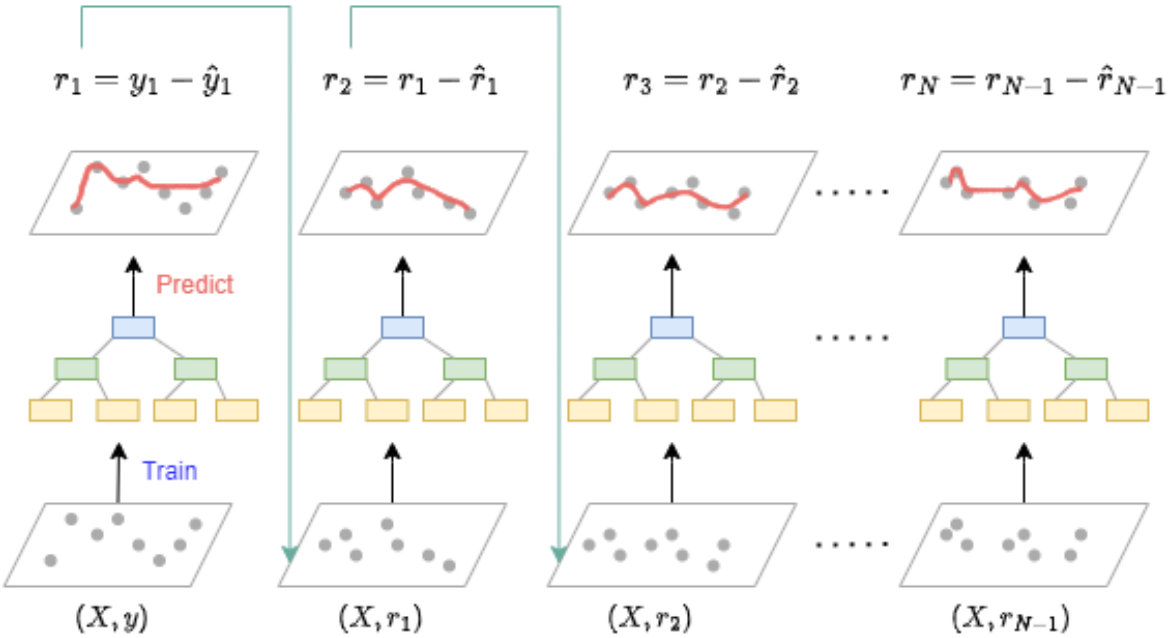


Figure 4.6: Depiction of gradient boosted trees trained for regression

For features X and a target y, XGBoost runs successive rounds of training, predicting, and incorporating the residuals r_n.

Extreme gradient boosting (XGB) implements gradient boosting with enhancements: it includes ℓ_1 and ℓ_2 regularisation to prevent overfitting, uses an efficient splitting algorithm, and also employs second-order gradient optimisation. It is widely used by data scientists as a popular tool for ML problems [10].

Random Forest

A Random Forest (RF) is an ensemble learning method that builds a large number of decision trees during the training process and then combines their outputs using a majority vote to make a prediction. By using voting and ensuring diverse trees through randomness, RFs mitigate the risk of overfitting, which is a common issue with more simple decision trees [3]. The randomness in an RF is twofold. Firstly, RFs use "bagging" where each decision tree in the training phase is built on a different random data subset. This helps make the trees more diverse to reduce overfitting. In addition to randomly selecting only some rows, RFs also randomly sample from the features at each split in the decision trees. This allows different trees to focus on varying aspects of the data, again helping ensure diversity and prevent overfitting, which is the key disadvantage of decision trees.

This concludes the contents of this Chapter, which sets out the four key dimensions of my modelling approach: the target variable, the feature variables, the observation sets, and finally the models. The results of these decisions follow now in Chapter 5.

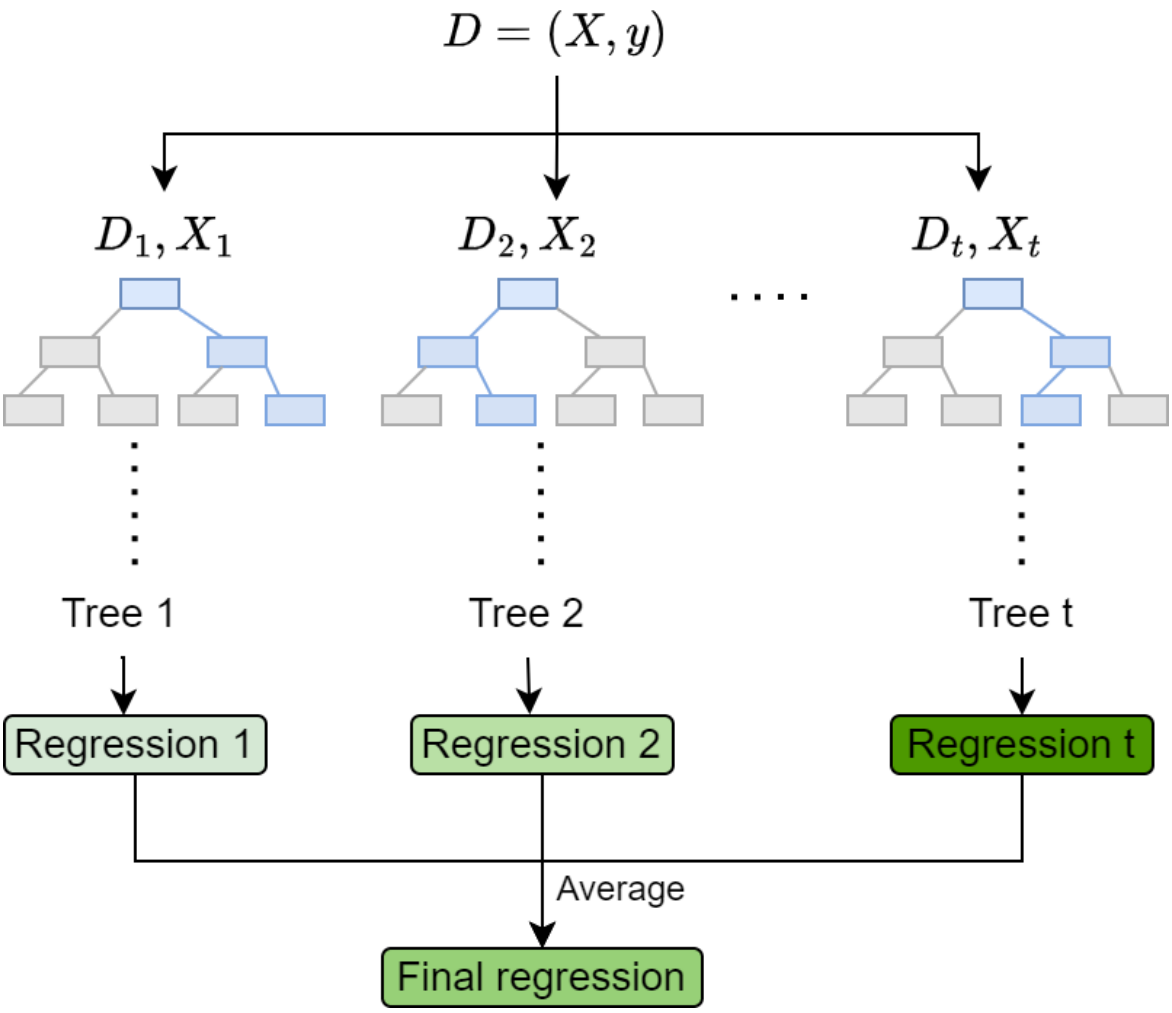


Figure 4.7: Depiction of a random forest trained for regression

For dataset D with features X and target y, I train trees on subsets of the data (bagging) and features.

Chapter 5

Results

As described in the previous Chapter, there are several important choices to be made: which variable to target, which explanatory variables to include, which observations to subset on, and finally which model to employ. To simplify this results Chapter, I begin with just focusing on the primary target variable of interest - predicting total S3 emissions. Once the logic has been set out for total S3 emissions, I will then discuss the results from modelling S3 upstream, S3 downstream, and S1 and S2 emissions.

5.1 Total S3 emission predictions

5.1.1 The target variable

The target variable can be in either absolute or intensity terms (see Section 4.1). To compare these options, I run all eight models across the three feature sets (extended, core, limited), and observation sets (all, Ex Covid, material). I plot the results on a set of graphs, with the x-axes showing the interquartile range of percentage errors, and the y-axes show the median percentage error. I draw an arrow from the absolute result to the intensity result. If an arrow points south-west, it implies that models that use intensity targets exhibit better predictive accuracy and lower volatility. An arrow pointing north-east implies the opposite. Any arrow pointing either north-west or south-east implies a trade-off between estimate accuracy and volatility.

From Figure 5.1, it appears that most arrows are pointing south-west. Therefore, target variables using carbon intensities generally perform better than those with absolute emissions. Accordingly, the first choice I make is to use intensity-based target variables.

5.1.2 The feature set

I repeat a similar exercise for the three feature sets. I estimate S3 total intensities using my “limited”, “core”, and “extended” feature sets. I plot the results showing performance in the median absolute error, and the variance in the IQR as before. In

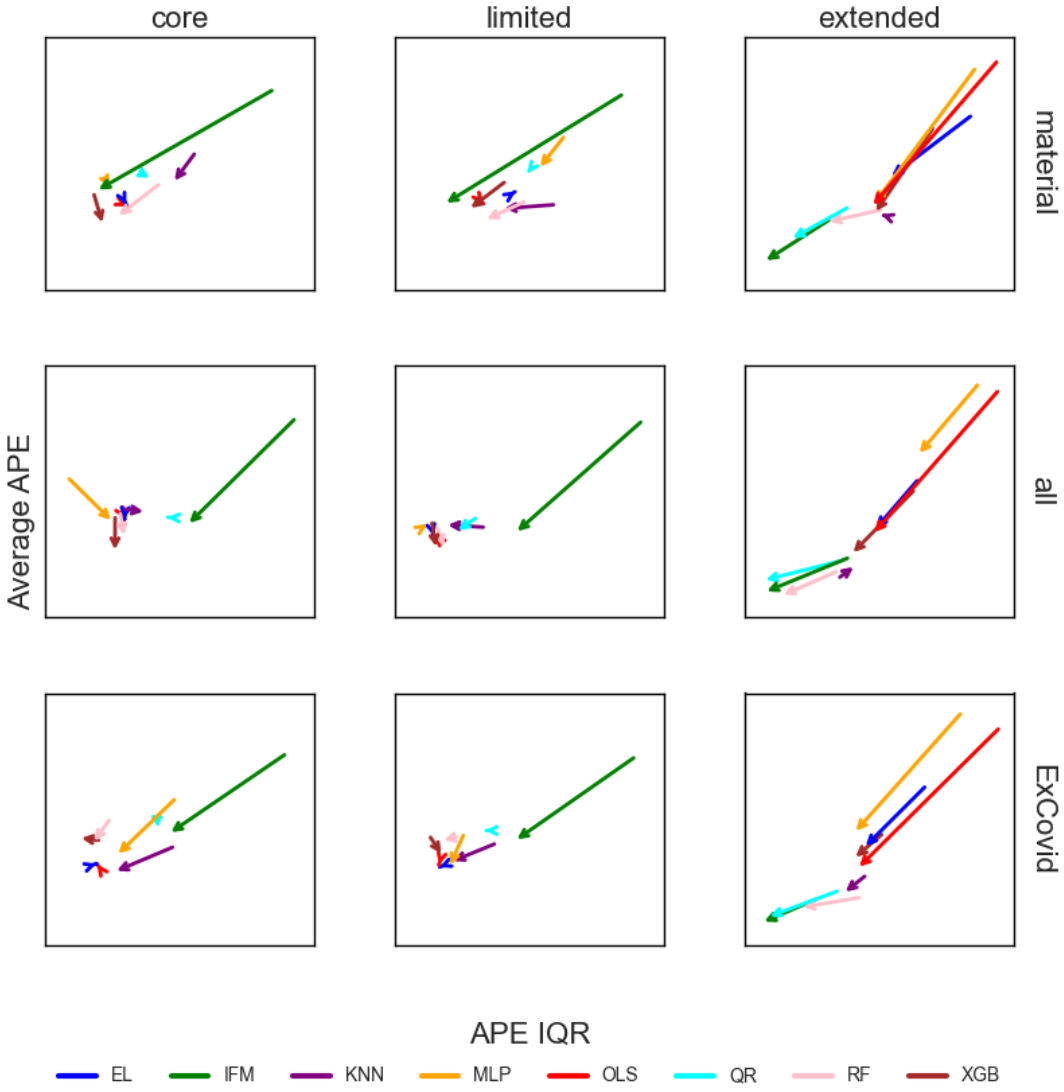


Figure 5.1: Absolute versus intensity modelling: S3 category total

The base of each arrow refers to the results from a model based on absolute emissions. The tip of each arrow corresponds to the results from a model based on emissions intensity. The arrow direction indicates how prediction accuracy and variability change depending on the target variable. Coloured arrows refer to the models specified in Section 4.4.2. The x-axes are the IQR of the absolute percentage error, and the y-axes are the median absolute percentage errors. The scales are not shown as only the direction of the arrow matters for these charts.

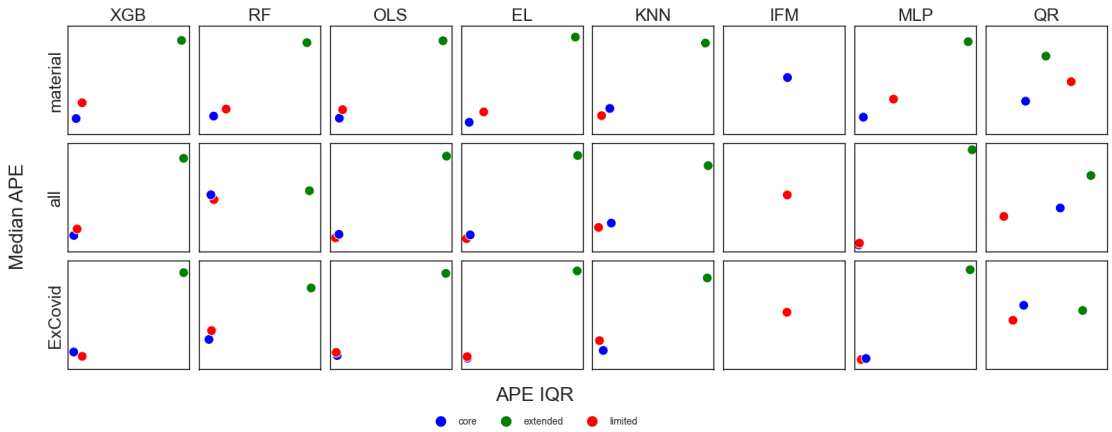


Figure 5.2: Comparing feature choice sets: S3 category total

Blue dots refer to the core set of features, green the extended set, red the limited set. Refer back to Section 4.2 for details.

general, I observe that the core (blue) and limited (red) set of features south-west of the extended (green) set of features. The exception is for the industry-fill model, where all points overlap because the choice of features is irrelevant. To proceed, I focus on using the core set of features. There doesn't appear to be any significant gain from using the limited set of features, and by including a few more variables, the explanatory power of the models may improve.

5.1.3 The observations

To assess which observations to include, I train each of the models and compare their performance using all observations, those excluding 2021 (to proxy omitting Covid), and those observations which are only material. As with the previous choices, I plot the percentage error IQR on the x-axis and the median absolute percentage error on the y-axis to understand the potential trade-off between prediction accuracy and variability. The results are given in Figure 5.3.

These results indicate that training on and predicting material observations produces the most accurate predictions. This also exhibits the lowest variation in IQR across the eight different models. Next, using all observations performs reasonably well for prediction accuracy but, depending on the model employed, shows a wide range of IQR. Finally, excluding Covid by omitting 2021 observations yields the least accurate predictions. This suggests that trying to eliminate the potential noise found in 2021 data throws out too many observations that are helpful to train the models.

Note that by using material observations only, the nature of the exercise shifts slightly. I am no longer trying to best predict what all firms report for their S3 GHG emissions. Rather, I am trying to model material disclosures. This means, for example, that my models will likely do a poor job at predicting total S3 emissions reported by a firm

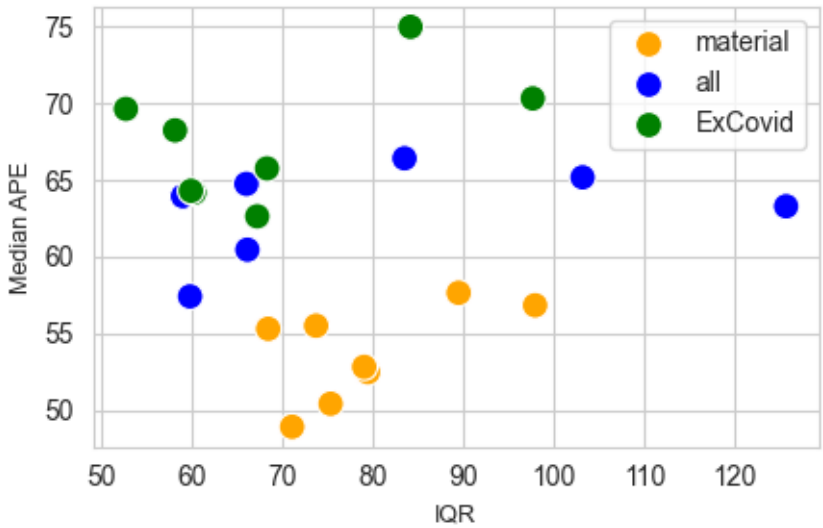


Figure 5.3: Comparing observation sets: S3 category total

Refer back to Section 4.3 for details.

that only reports a minor category, such as employee commuting.[1] However, this is arguably a better outcome. By including all observations, the models are trained on observations where firms are under-reporting; the focus on materiality ensures that the majority a firm's S3 emissions are always being considered (see discussion in Section 6.2).

5.1.4 The models

Having decided to model total S3 emission intensity on the core set of variables using material observations, it remains to compare the relative performance of the eight models described in Section 4.4.2.

Measuring model performance

Regarding performance measures, I largely focus on absolute percentage errors. I care about absolute errors as I value under- and overestimates symmetrically. Using percentage errors is helpful because they are easily interpretable and they normalise the error.

The primary metric I use to measure model prediction performance is the median absolute percentage error (MDAPE).

$$\text{MDAPE} = \text{median}\left(\left|\frac{y_1 - \hat{y}_1}{y_1}\right| \times 100, \left|\frac{y_2 - \hat{y}_2}{y_2}\right| \times 100, \ldots, \left|\frac{y_n - \hat{y}_n}{y_n}\right| \times 100\right) \quad (5.1)$$

[1]This is visualised in Figure 6.5.

I use MDAPE because it is a robust way to assess percentage error, being less sensitive than the mean. Nevertheless, the spread of estimated errors are also important, and so I also measure the interquartile range (IQR) of the absolute percentage errors (APE).

$$\text{IQR(APE)} = Q_3\left(\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\right) - Q_1\left(\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100\right) \tag{5.2}$$

where Q_3 is the third quartile (75th percentile) of the absolute percentage errors, and Q_1 is the first quartile (25th percentile) of the absolute percentage errors.

Following Nguyen et al. [48], I also measure the proportion of estimates that lie within $\pm 50\%$ of the true value. In accordance with their terminology, I refer to this measure as the "proportion of prediction in acceptance range" (PPAR):

$$\text{PPAR} = \frac{1}{n} \times \sum\left(1 \text{ if } \frac{y_i - \hat{y}_i}{y_i} \leq 50\%, 0 \text{ if otherwise}\right) \times 100 \tag{5.3}$$

The final metric that may be of potential interest is the model's R^2. This measures how well the model fits the data, providing the ratio of variance in the target that is explained by the dependent variables.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{5.4}$$

Despite its name, the R^2 can be negative if the model performs worse than simply predicting the mean of the target for all observations. This happens if the residual sum of squares is larger than the total sum of squares (the variance of the data around the mean). In other words, the model would perform worse than a horizontal line (representing the mean) for predicting the target.

Model comparison

Table 5.1 summarises the performance of my models, providing several important insights. First, in terms of prediction accuracy, tree-based methods appear to perform the best with the lowest MDAPE and highest PPAR. However, even these techniques produce only modest improvements over the naive baselines. This accords with the conclusions of Nguyen et al. [48] who also find that absolute prediction performance is low, even with the best models, and that model performance is broadly similar across a range of models varying in complexity.

Nevertheless, my models represent a meaningful improvement. Nguyen et al.'s best performing model yields a MDAPE of 72.23 percent and a PPAR of 30.36 percent.[2] In

[2] A stepwise OLS using the average of 5-fold cross validation on S3 emissions aggregated across the fifteen categories.

Table 5.1: Model performance for predicting S3 cat total emissions

	MDAPE	**IQR_APE**	**PPAR**	**R-squared**
IFM	55.31	68.41	45.65	0.16
OLS	52.49	79.44	47.39	0.11
QR	57.66	89.48	39.57	0.06
EL	52.83	79.03	46.52	0.11
MLP	55.51	73.71	43.48	0.10
KNN	56.85	97.93	45.22	-0.01
XGB	48.92	71.06	50.87	-0.04
RF	50.42	75.32	49.57	0.06

Predicting S3 cat total intensity using the core feature set over material observations. Green cells denote the best performance in that respective metric over all models.

particular, my XGB model represents a significant improvement on both these scores. As Nguyen et al. [48] use a different data source and sample selection, it is difficult to unambiguously assert that my approach of using intensities, fewer explanatory variables, and material observations is the superior approach. Still, it is encouraging that my results show an improvement on the only other S3 ML paper.

Considering the other metrics, simpler models exhibit a higher R^2 (IFM, OLS, EL) and IFM also has the lowest APE IQR. By contrast, while XGB has the best predictive accuracy it has a very low R^2. There are several reasons why XGB might do well in one metric and poorly in another. I briefly consider the two most plausible reasons:

- Nature of the data: If the data has a lot of inherent noise, XGB can perform well in terms of predictions but might not capture variance effectively, leading to a low R^2. Given the concerns around data quality and coverage discussed in earlier sections, this is the most plausible explanation.

- Model complexity: If S3 emissions have high variance and this isn't explained well by the features, my XGB model might provide accurate predictions for certain instances, but struggle to explain the overall variance. This could be true in this case, and there does appear to be a weak (inverse) relationship between model complexity and R^2.

Other reasons why the R^2 could be low include the potential of overfitting, poor scaling/transformations, and the case where the simple mean baseline explains a significant portion of the variance well. I discount the first two of these due to my modelling process (refer to Section 4.4.1). The last reason seems unlikely due to the poor performance of the IFM model which also relies on a simple mean-based estimate.

Looking at the values of the metrics in Table 5.1, together these suggest that ML techniques do not yet provide a clearly superior approach over naive methods. This is somewhat in line with the previous study by Nguyen et al [48], who find that

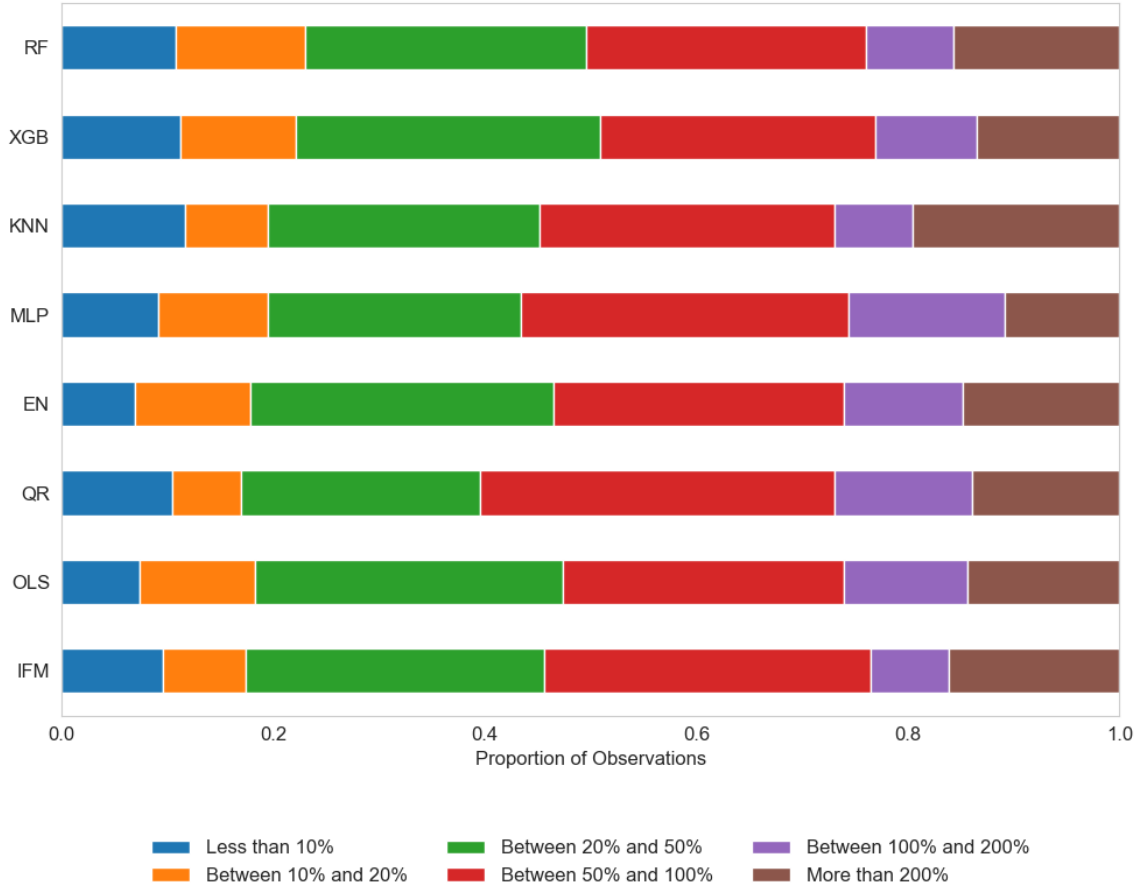


Figure 5.4: Model prediction accuracy bins for total S3 emission intensity

Inspired by [23]. Predictions made for S3 category total intensity, using the core set of features over material observations.

even the most advanced ML techniques produce only very modest improvements in prediction accuracy for estimating total S3 emissions.

To characterise the effectiveness of each model's predictive performance further, I bucket the predictions into ranges and plot them in Figure 5.4. This also serves to highlight the surprising similarity in performance across the range of models. While tree-based methods offer some improvement over the baselines, such improvements do not represent step-changes in prediction performance.

The limitations in absolute prediction accuracy are also in line with the report by FTSE-Russell as shown in Figure 5.5. Whichever approach they take, the find that prediction accuracy is low, even for scopes 1 and 2. However, my models appear to do better in reducing the share of large outliers, i.e. errors greater than 100 percent.

Together, Table 5.1 and Figure 5.4 imply that ML approaches provide, at best, only a modest improvement in predictive accuracy. To illustrate this further, in Figure 5.6

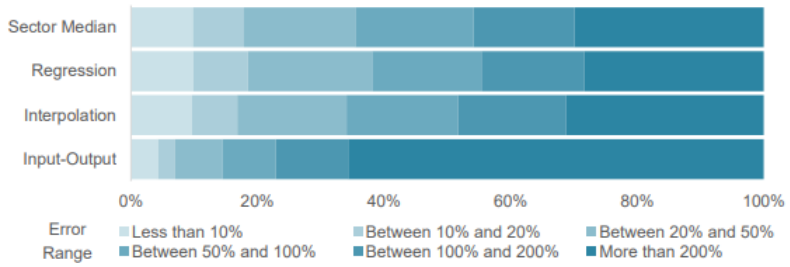


Figure 5.5: Share of estimated observations within error thresholds by model, S1 and S2

Source: FTSE Russell, [23]

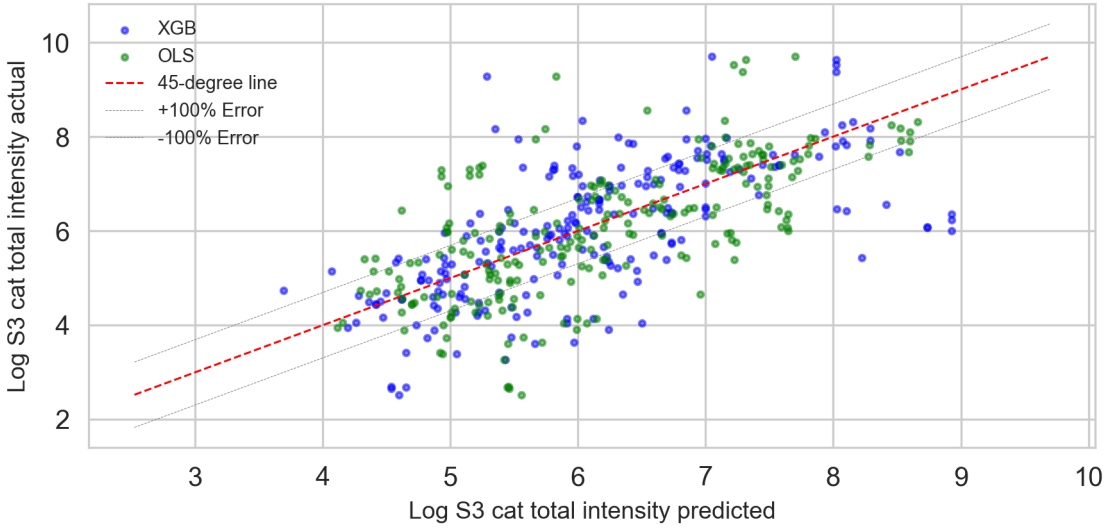


Figure 5.6: XGB and OLS predictions vs actual values

Predicted log emission intensities plotted against observed log emission intensities for S3 category totals, core explanatory features, material observations.

I plot model predictions versus true values. Points along the 45 degree line show an exact match. Using log-scales, any point within the dashed lines is within 100 percent APE. XGB points are given in blue, and OLS in green.

While there is clearly some difference between XGB and OLS, the general trend in the data remains the same, implying that the predictive difference between these models is modest. Furthermore, OLS is much more tractable for transparency and explainability reasons, and so in the context of a commercial product, this may be the preferred option (at least until ML performance improves significantly).

To summarise the findings for total S3 emissions, I use an iterative approach to narrow down my choices. Guided by like-for-like comparisons of MDAPE and IQR, I find that estimating S3 intensities produces better results than absolute S3 levels. In a similar fashion, using a core set of explanatory variables seems to be a good

compromise between the extended and limited feature sets. Third, predicting with material emissions significantly improves predictive accuracy (as per Figure 5.3), and also produces estimates that guard against under-reporting, consistent with the precautionary principle. Finally, tree-based models (especially XGB) perform the best for predictive accuracy, but gains over simple baselines are only marginal.

5.2 Upstream, downstream, and S1 and S2 results

For the sake of brevity, rather than repeat each step of the analysis in full for the other three target variables, I shall instead present the key findings. To begin with, after whittling down the choices using the aforementioned experimental approach, Table 5.2 present the model metrics following the format of Table 5.1.

Table 5.2: Model performance for predicting different emissions

Target	Model	MDAPE	IQR_APE	PPAR	R-squared
S3 upstream	IFM	41.94	55.13	59.13	0.34
	OLS	48.27	45.99	53.04	0.42
	QR	50.72	52.44	49.13	0.42
	EL	48.99	46.56	51.30	0.40
	MLP	55.70	48.10	43.04	0.36
	KNN	49.83	45.57	50.00	0.26
	XGB	49.93	50.46	50.00	0.31
	RF	47.34	47.27	53.91	0.29
S3 downstream	IFM	73.82	152.73	28.96	0.18
	OLS	69.12	70.16	35.75	0.14
	QR	76.30	114.77	35.29	0.06
	EL	68.57	66.13	34.84	0.13
	MLP	66.62	63.25	38.01	0.07
	KNN	70.62	83.79	37.10	-0.00
	XGB	72.98	68.08	31.22	0.07
	RF	72.58	60.16	28.05	0.08
S1 and S2	IFM	63.09	74.04	41.73	0.17
	OLS	61.14	81.70	41.03	0.18
	QR	69.90	104.14	37.90	0.07
	EL	61.20	81.04	40.93	0.18
	MLP	58.31	67.97	44.86	0.13
	KNN	64.55	102.39	38.61	0.16
	XGB	59.88	73.16	43.55	0.21
	RF	60.34	80.85	41.13	0.10

Best combinations are as follows: S3 upstream is modelling intensity over the core set of features for material observations. For downstream, it is modelling intensity over the core set of features for all observations. This is also true for S1 and S2 emissions.

As with the S3 total, the best combination for modelling S3 upstream emissions is to model intensity over the core set of features for material observations. The

best model for MDAPE and PPAR is the IFM which significantly outperforms the other models. However, the IFM also has the largest average spread in its estimates (55.13). Moreover, the models' R^2 is respectable and much higher than the case for S3 total emission intensity.

Downstream emissions are best modelled using intensities over the core feature set, but using all observations instead of material observations. This could be because there are far fewer downstream observations to begin with, and so reducing the number still further using the materiality mask results in worse performance (see Figure 6.6 in Section 6.3 for a count of observations for each S3 category). For the model, the MLP produces the most accurate predictions, but only marginally so (e.g. compare against KNN). Improvements over the baselines are largely in respect of prediction variability, with most other models exhibiting lower IQR APE. In general, it appears that S3 upstream produces the best results, S3 downstream the worst, and S3 total - being comprised of both - a middle ground.

Surprisingly, the best results for S1 and S2 modelling are notably worse than for both S3 total and S3 upstream modelling. This is despite S1 and S2 observations being over three times that of S3 (see Table C.1), and the general acknowledgement that S3 data is worse quality than either S1 or S2. However, unlike previous studies, I subset my S3 observations to only include material data. By contrast, for S1 and S2 data, as there are no categorical breakdowns or materiality conditions, all data points are used. I believe that this adds a nuance to the modelling debate: in general, S1 and S2 data have better coverage and quality than S3 data; however, S3 material observations can compensate for the lack of data quality and, so, result in superior predictive performance over S1 and S2 emissions.

To conclude this Chapter, when reviewing how different models perform across the combinations of target variables, features, and observation sets, no single model has proven unambiguously dominant: there is no pattern in Table 5.2. Moreover, in many cases, model performance is broadly similar across the range of metrics. In addition, where clear improvements have been evidenced (e.g. in Figures 5.1 and 5.3), this has been driven by refining the data used for the modelling rather than the choice of model itself. In sum, these results imply that improvements to S3 modelling are best achieved through careful choices related to the input data rather than selecting a superior modelling technique.

Chapter 6

Applications and extensions

Having set out the main results in the previous Chapter, here I will discuss three further applications and extensions of my analysis. Focusing on total S3 emission predictions, first I will explore the outliers identified by the modelling. Second, I will apply the material estimation model to try to gauge under-reporting. Finally, I will also try to model category one S3 emissions (purchased goods and services) and compare the results against the findings from the aggregated target variables.

6.1 Outlier analysis

Figure 6.1 uses the predictions from the XGB model described previously in Table 5.1. Firms A and B (orange and purple points respectively) show where the model significantly overestimates the true S3 emission intensity. By contrast, Firm C (green) represents an instance where the model underestimates the true value.

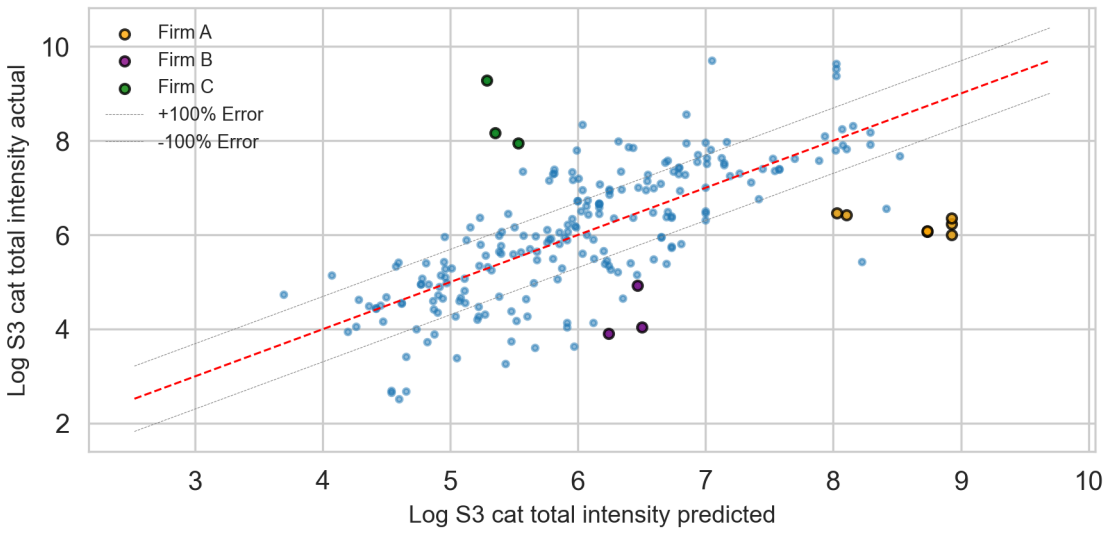


Figure 6.1: XGB predicted vs actual S3 total emissions

6.1.1 Firm A

Firm A is Lockheed Martin Co. (LMC), a global security and aerospace company employing over 100,000 people. My XGB model significantly overestimates is true S3 intensity. As is shown by Figure 6.2, when compared against other firms that report material emissions in its peer group, it reports an S3 intensity orders of magnitude lower than other firms of similar size.

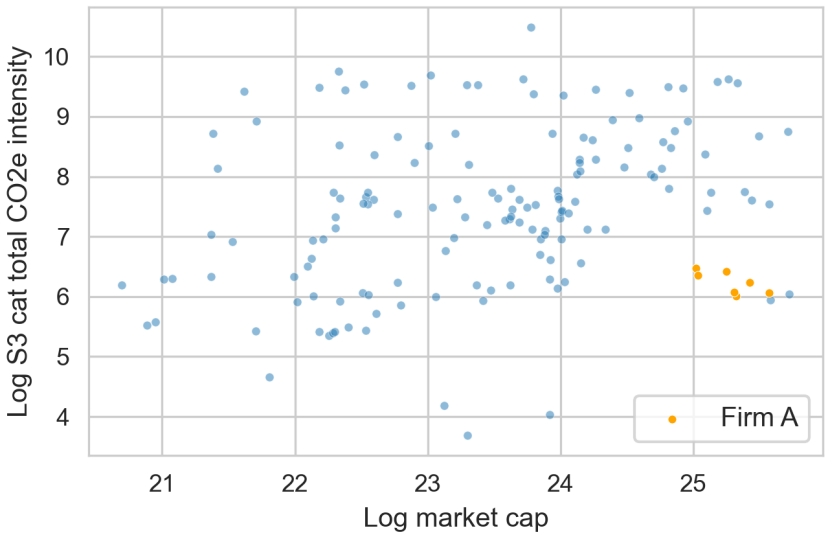


Figure 6.2: LMC (Firm A) peer group: S3 intensity vs mcap

After reviewing the history of its S3 disclosure, one plausible explanation is that LMC is one of the furthest along in its efforts to measure and reduce S3 emissions in its peer group. They were one of the earliest firms in the aerospace and defence industry to perform a high-level economic input-output life cycle analysis to track environmental impacts throughout the supply chain. This analysis was completed and reported for the first time in their 2012 CDP response. Moreover, in the same year, the firm began detailed estimations and disclosure of S3 emissions [41].

These efforts have since continued. To strengthen the quality of their S3 emission calculations, they plan to increase primary emissions data collection, setting a target to implement a third-party validated supplier sustainability assessment programme by 2025, which would include outreach to suppliers representing 60 percent of their spend. They also contribute to industry-led supply chain resources and partnerships, and invest in a set of R&D projects to spur innovation and reduce emissions further. Finally, according to their latest carbon reduction plan for the United Kingdom, they are far exceeding their targets against their 2019 baseline [42].

Together, this suggest that LMC is among the most advanced reporters for its peer group. Therefore, it is possible that the model performs poorly because LMC is doing relatively better than what is to be expected of firms with a similar profile.

6.1.2 Firm B

Firm B is Mitsubishi Corp (MC), a Japan-based company mainly engaged in living, mobility infrastructure, energy, electric power and others industrial fields. It is a very large conglomerate comprising ten business segments. As with LMC, my XGB model significantly overestimates the reported S3 emission intensity for MC. To understand the numbers in more detail, Figure 6.3 shows the raw data loaded from the API for MC's category emissions. Here, the issue appears to stem from methodological changes resulting in volatile reporting.

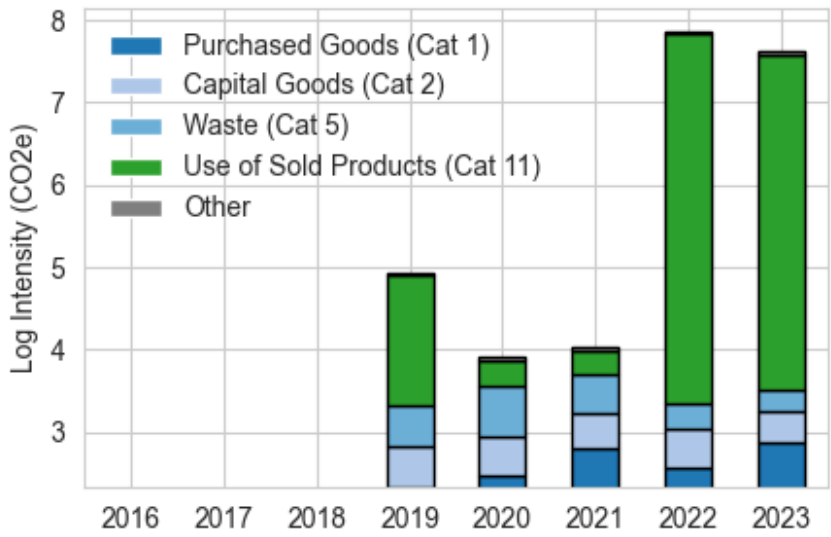


Figure 6.3: MC's (Firm B) S3 category emissions

Before 2019, no data is reported. Then, the years 2019-2021 provide a breakdown of the three points highlighted on Figure 5.6. The drop from 2019 is largely due to a reduction in reported category 11 emissions.

The years 2022 and 2023 show a dramatic increase in S3 emissions, but this data is not incorporated into my model. The year 2023 is out of scope as, due to reporting lags, much of the data is yet to come in for other firms. To prevent modelling 2023 on a partial sample, I choose to only model up to the year 2022.

Regarding MC's 2022 numbers, the downstream emissions data is actually already winsorized by LSEG's internal processing rather than representing the true reported value from the firm. As my model only takes into account reported values directly from firms, MC's downstream categories are masked; as this includes category 11 which is a material category for MC's peer group, this results in MC containing no data for my model in 2022, and so Figure 5.6 only shows three data points. Were the 2022 and 2023 to be plotted on Figure 5.6, the points would sit very close to the +100% error line, with MC going from an overestimate to an underestimate. This highlights how my stringency in only using firms' reported data can lower coverage.

Earlier in Section 2.4, I discussed how issues in the data generation process led to knock-on problems with the data; here, we see how MC provides a real-world example of reporting inconsistency. In MC's 2022 Sustainability Report, they report "following GHG emission reduction targets announced in FY2021, [MC] calculates GHG emissions based on the GHG Protocol's equity share approach from the year ended March 2022" [47].

To collect the evidence together, first, it appears that before 2019, MC didn't report at all. The years 2019-2021 represent a partial disclosure as the change in category 11 in these years suggests a change in method. Years 2022 onwards then represent another change to methodology using the GHG Protocol's equity share approach. Together, this results in highly volatile data that is difficult to compare year-on-year. Therefore, it is perhaps unsurprising that even my best model yields large prediction errors for MC.

6.1.3 Firm C

In contrast to LMC and MC, Firm C's predicted S3 intensity is significantly below its reported S3 intensity. Firm C is Fluidra SA (FSA), a Spain-based company active in the pool and wellness equipment business; its business sector (peer group) is cyclical consumer products. Figure 6.4 situates FSA's total S3 intensity against the other firms in its peer group.

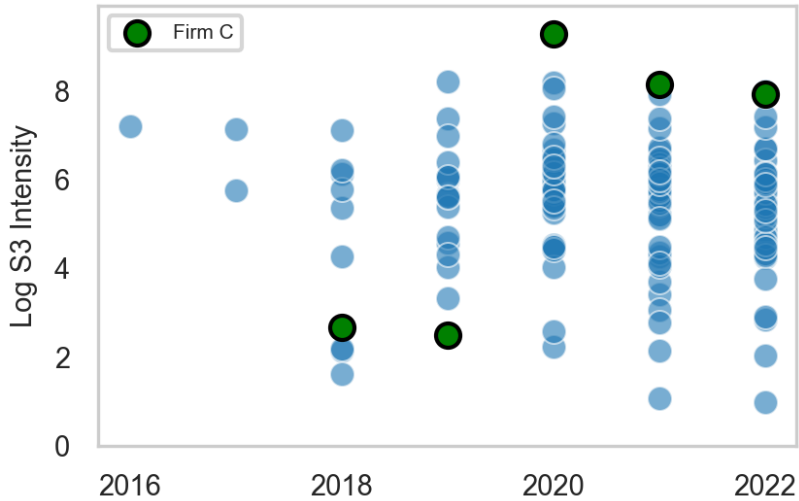


Figure 6.4: Cyclical consumer product S3 intensities, full sample

As with MC, FSA exhibits activity exclusion. In 2018 and 2019 I have no downstream categorical data. In my code this sets downstream emissions to zero, and so the 2018 and 2019 data points simply reflect the total reported upstream emission intensity. However, this is effectively dealt with using material emissions only: the materiality

mask removes these data points from the modelling process, ensuring I don't train models on under-reported data such as these.

From 2020 onwards, a more complete picture of S3 categorical data emerges, and as is shown by Figure 6.4 FSA consistently reports the largest S3 emission intensity within its peer group. At least part of the explanation might be in its business: heated pools are among the most energy-intensive consumer products on the market.

However, another reason relates to the boundary incompleteness and difficulties with pulling together accurate data throughout the supply chain. For example, FSA operates in over 45 countries and this has had knock-on methodological impacts. According to their annual report, while beforehand they used only one emissions factor, "in 2021 the IEA emission factor corresponding to the country of sale of the product has been considered." They add that "looking ahead to 2022, [FSA] will continue to integrate the emissions associated with the energy consumption of its products during their useful life, until the entire catalog is covered" [20]. Therefore, from 2021 FSA moves from one emission factor to dozens, as well as expanding the catalogue of products included in its reporting. In other words, what they report and how evolves significantly.

For the 2022 data, FSA explains the continued high variation in their underlying S3 data from sources such as: "an increase or decrease in the corresponding activity (as is the case for categories 2 and 5), an increase in the scope of the information collected (categories 2, 4, 6, 9 and 11), or the integration of methodological adjustments (or new emission factors) in their calculation (categories 1, 2, 3, 7, and 9)" [21]. In this case, when considering the variation in underlying data, it is difficult to disentangle the "truth" (changes in corresponding activity) from the data quality or coverage, especially with an aggregated target variable such as total S3 intensity.

This outlier analysis provides real-world instances of the potential issues discussed earlier in Section 2.4.1. While activity exclusion can largely be dealt with using a material categories filter, for some firms the model may struggle due to reporting inconsistency and/or boundary incompleteness. This is a useful reminder that I am modelling a wide portfolio of firms that vary in their S3 reporting sophistication. Some firms only partially report and are screened out by materiality. However, even for those reporting material categories, methodologies can often begin crudely and improve over time, along with more refined data collection practices.[1]

Furthermore, for some firms, the model may struggle if they are at the frontier of S3 efforts, not only providing detailed and third-party verified estimates, but also actively engaging with suppliers and undertaking meaningful initiatives to reduce their carbon footprint.

In each case, the model nevertheless provides a useful first step. Where predictions

[1]Incorporating methodological information will be discussed further in Section 7.3.

deviate far from the reported values (visualised using the perpendicular distance from the red line in Figure 6.1), as these examples illustrate, this may reflect how far along the firm is in their S3 reporting methodology.

6.2 The precautionary principle

The precautionary principle concerns how policy-makers act in high-stakes settings with large uncertainties. In simple terms, it tries to deter inaction due to uncertainty when such inaction has potentially catastrophic and irreversible consequences. It is often invoked in the areas of human health and the environment. For example, the precautionary principle might suggest that just because there is high uncertainty in climate projections, this doesn't imply that doing nothing is the best approach.

In the context of carbon reduction targets, it remains an open question as to how to apply the precautionary principle. The key problem is that while there are binding targets, many firms do not report their emissions. Therefore, how can one undertake S3 estimation given the need to incorporate and overcome this uncertainty?

In their paper, Hoepner and Rogelj [32] argue that researchers need to assume that non-reporting firms or under-reporters are worst-in-class. As a consequence, their GHG emissions should be consistently and precautionarily overestimated at the 95th, 99th, or 99.9th percentile of their relevant peer group. The main aim of their approach is to incentivise non-reporters to disclose and encourage under-reporters to improve their accuracy.

As a simple exercise, I compare the median S3 total intensity to the 95th percentile for the firms in my sample. If I group by their economic sector, the ratio is 8.3 (i.e. the 95th percentile amount is 8.3 times the reported median), and if I group by their business sector (one level down) the ratio is 8.9. Therefore, were I to apply this to non-reporting firms, I would be estimating GHG emission intensities of between 8 and 9 times the peer group median on average.

While such an approach would penalise non-reporters and perhaps incentivise them to disclose, as noted by Olesiewicz and Kooroshy [50] this approach is impractical for corporate emissions estimates, given the number of non-reporting companies, high degrees of firm heterogeneity and unrealistically high estimates which also tend to be volatile year-on-year.

My framework offers an alternative way. As mentioned previously in Section 5.1.3, by training my models on material S3 emissions I guard against the under-reporters influencing the models' learning. Using a model trained in this way, Figure 6.5 shows how this can apply to non-material observations. Specifically, I divide my dataset into material and non-material observations according to their business sector. Next, I train an XGB model using material observations, and I make predictions on only the non-material observations.

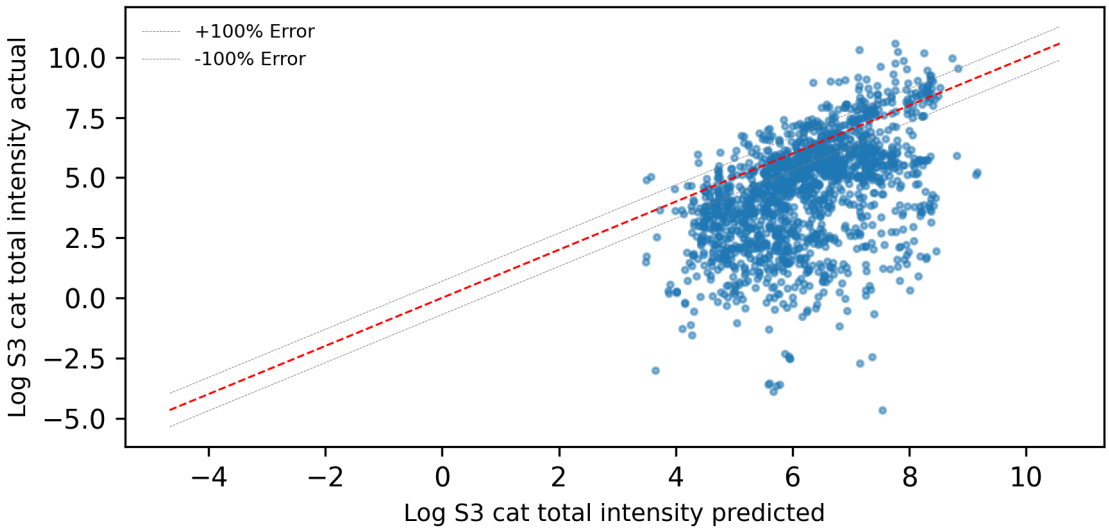


Figure 6.5: Predicted S3 intensity for non-material observations

Note: XGB model trained on material S3 observations and fitted to non-material S3 observations above.

From Figure 6.5 it is clear to see that in most cases, fitting non-material observations by a model trained on material observations results in reported intensities below their predicted value, as one might intuit. Nevertheless, there are also some points above the red-dashed line, i.e. their reported values are higher than their predicted (material) values. This is a useful reminder that materiality does a very good job *on average* of guarding against under-reporting, but it is not perfect for every firm.

A precautionary principle applied to these under-reporters could be the maximum of either their reported S3 total intensity, or their predicted S3 total intensity based on the model trained on material observations. This approach both guards against under-reporting while also respecting the cases where reported values are above what would be expected. As required, this would provide a consistent precautionary approach but without estimating values that are unrealistically high.

6.3 Predicting a S3 category

As described in Section 3.2.1, I have chosen not to try to model each S3 category separately due to the paucity of data in some categories. Rather, I have opted to aggregate categories into either upstream, downstream, or total S3 emissions.

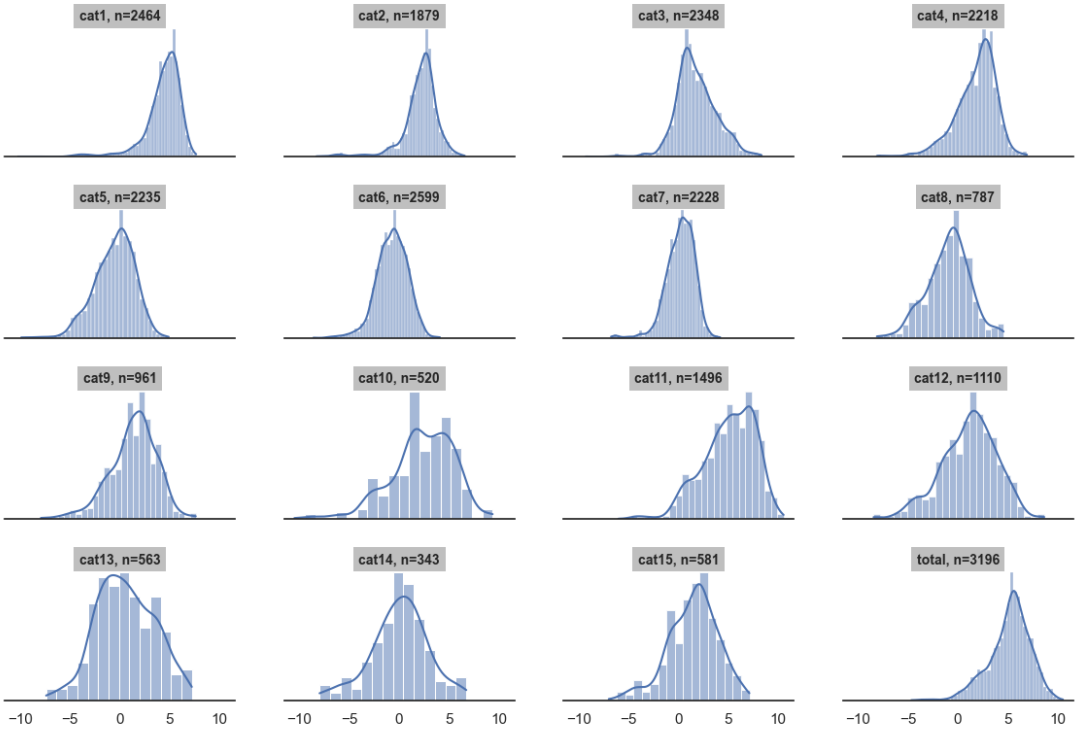


Figure 6.6: Distributions of S3 intensities by category

Distributions based on all available data points in the sample.

Notwithstanding this, Figure 6.6, shows that there are some relatively well-covered categories.[2] In this Section, I have chosen to try to model category one (purchased goods and services) emission intensity and compare this against the results for total S3 emission intensity.

I select category one for this exercise because, as shown by Figure 3.2b and Table B.1, category one is material for almost every peer group and relatively well-disclosed. Accordingly, it provides me with the best mix of coverage, quality, and relevance. Together, this perhaps provides a picture of what data quality and coverage could look like for other categories in the future if they were able to catch up with scope one's levels of disclosure and consistency. This makes it the ideal candidate for an extension to the main body of my analysis, hinting at what might be possible for the future of categorical modelling.

I model S3 category one intensity using all available observations in my sample; I split the data into eight folds with a minimum of three firms per peer group. One fold is randomly selected as the holdout test set and cross-validation is performed

[2]Note the difference in coverage and variance between upstream and downstream categories. This may go some way to explain the poorer results of modelling downstream emissions discussed in Section 5.2.

Table 6.1: Model performance for predicting S3 cat one: purchased goods and services

	MDAPE	**IQR_APE**	**PPAR**	**R-squared**
IFM	49.86	65.46	50.16	0.29
OLS	56.19	59.61	44.51	0.24
QR	56.53	95.22	42.95	0.26
EL	56.01	56.52	44.83	0.24
MLP	56.48	44.36	42.32	0.14
KNN	56.07	56.98	42.95	0.13
XGB	53.01	61.52	47.65	0.20
RF	54.50	58.60	45.77	0.22

Predicting S3 category one (purchased goods and services) using the core feature set over all observations. Metrics are defined according to Section 5.1.4 above. Green cells denote the best performance in that respective metric over all models.

on the remaining 7 folds for each model. The results are given in Table 6.1.

The magnitude of prediction errors (MDAPE, PPAR) and ranges (IQR APE) are broadly in line with the results for S3 category total intensity modelling, summarised in Table 5.1. However, there are two interesting differences. First, the R^2 is notably higher for all models, implying that models have more explanatory power for this individual category compared to the aggregated categories. Second, unlike in the case of S3 total intensity where XGB was on balance the best model, in the case of category one the IFM appears superior. It is also worth noting that my results are superior to those in Nguyen et al. [48], who report an MDAPE for category one of 81.59 and a PPAR of 22.86. However, as previously mentioned, due to the difference in data sources and sample construction, caution should be applied when trying to compare these results.

As noted by other studies, a firm's sector can have an important bearing on its mix of upstream and downstream emissions because different sectors are at different points along the supply chain [25]. For example, as energy producers and industrial firms provide inputs to production, they typically exhibit a greater share of downstream emissions than other sectors. Due to this, it makes sense that the firm's peer group may be even more important for modelling one category than when modelling total S3 emissions. Therefore, it is not unbelievable that the simple IFM might provide the best results on a category-by-category basis.

To summarise this brief exploration into modelling a single category - category one - the broad measures of prediction accuracy and range are comparable to modelling total S3 GHG emission intensity. However, the explanatory power of the models is considerably higher, and the best-performing model is now the simple baseline IFM rather than the tree-based methods. This supports my earlier conclusion that, unless there is a step-change to the data quality and coverage of reported S3 emissions, the utility of ML methods over simple baselines is likely to be limited.

Chapter 7

Potential future research

Following the results given in Chapter 5, one might be tempted to conclude that the use of data science, AI and ML in the context of S3 emissions estimation is of dubious benefit. However, just because ML methods currently struggle to accurately predict S3 emissions, there remain promising avenues to explore. In this Chapter, I briefly set out a few examples which I think look encouraging.

7.1 Data collection, automation and interpolation

To focus the scope of this report, I have treated the categorical S3 reported data extracted from LSEG as my ground truth, and performed very limited interpolation on other explanatory variables (see Section 3.3 for details). This has the advantage of only using data directly reported by firms, but the drawback that it significantly reduces sample size given how many firms don't report at all. It may also introduce selection bias into the sample if certain types of firms (e.g. SMEs) are more likely under-report.

In a recent paper by Cullen, Marinoni and Cullen [13], to tackle the problem of incompleteness in GHG datasets arising from both inconsistent reporting and poor transparency, the authors try a range of ML methods to automate the completion of GHG datasets. They explore three gap-filling tasks of increasing complexity, with the aim of correctly classifying emitters into four buckets ranging from low to high. Like my approach, their techniques also display a range of complexity, including simple interpolation, shallow learning models, ensemble models, deep learning models, and graph representation learning models.

The authors find that for simple gap-filling problems and datasets with fewer input features, simple models outperform complicated models in output average accuracy. However, as the gap-filling problems become more complex and datasets comprise an increasing number of input features from which models can learn, more complex deep-learning models output the best average accuracy scores.

Cullen, Marinoni and Cullen show that while ML approaches are not the best way

to automate the solution in every case, learning-based classifiers can sometimes be used to effectively and scalably complete emissions datasets.[1] This helps address a key drawback to manual data-collection which is expensive, lacks comparison over different methodologies, and can be subject to human error.

7.2 Improving supply chain visibility

Another promising collection of research uses ML techniques to combine datasets together for new applications. In this Section, I refer to two recent papers which employ ML to help build data across the supply chain with a view of improving data availability for researchers, firms, and investors interested in S3 estimation.

First, Jain et al. [37] estimate S3 emissions by classifying transaction descriptions of purchased goods and services into the US Environmentally-Extended Input-Output (USEEIO) commodities. They can then map these transactions onto commodities purchased and, by using the appropriate emissions factors, come to an estimated carbon footprint. By using fine-tuned natural language processing (NLP) models, the authors demonstrate that new insights can be gained by intelligently combining two previously unrelated datasets.

Another example of a novel bottom-up approach is in a paper by Mejia and Kajikawa [46] who employ network analysis to estimate investment-associated S3 emissions. By building a shareholder network of Japanese public firms, they are able to allocate and estimate S3 emissions arising from investments.

What is common to both of these papers is the use of ML to improve visibility of S3 emissions sources throughout the value chain. They represent ways to disentangle complex interactions in the supply chain which can sometimes cloud the availability or reliability of S3 data.

7.3 Incorporating methodology

From Section 6.1, it is clear that accounting for the reporting methodology of firms is a crucial step to interpreting predictions (and, in my case, model outliers). As discussed earlier in Section 2.4.1, the latitude that the GHG Protocol affords firms means that, when aggregating up across firms (or even comparing within firms across the years), figures are often incomparable without qualification.

Accordingly, a useful approach would be to incorporate, perhaps as metadata, some key parameters about how the firm calculated their estimated S3 emissions. Such information might include the base year, key assumptions, whether they adopt an

[1] The authors acknowledge that there are limitations to their approach: their results are based on a four-class classification which is too imprecise for many applications, and uncertainty isn't quantified. They aim to address these shortcomings in future work.

ownership or control approach, how each category has been modelled, and so on.

The potential of this information is best-illustrated by considering the materiality mask. This has proven an effective strategy for removing lower quality observations, enhancing the predictive accuracy of models and implementing the precautionary principle. This has been straightforward to implement using simple rules to check which categories are reported or not. It would be a more difficult exercise to use large language models (LLMs) with retreival augmented generation (RAG) to tag the data with methodological details. However, doing so would likely add useful colour to data quality and, in a similar way to the materiality mask, give the analyst the ability to experiment with filtering observation subsets for determining the best trade-off between data coverage and quality.

An example of such an application could be to tag the reported values according to whether they have been compiled in accordance with the GHG Protocol or the CDP questionnaire. This would ensure that models could then discriminate between different methodological frameworks.

Estimating S3 emissions remains a promising field for applied ML. Following my analysis presented in the report, I believe that significant gains in predicting S3 emissions will be reliant on improved data coverage and quality, rather than in the elusive promise of a better model architecture. Accordingly, in this Chapter I have set out three avenues - automating interpolation, ML-generated datasets, and LLM RAG of methodologies - which may prove fertile ground for future progress.

Chapter 8

Conclusion

Given the environmental implications of S3, and the legal and social imperative for firms to reduce their carbon footprints, estimating corporate GHG emissions is an ideal topic for research. Nevertheless, the nature of S3 emissions means that data is scarce, costly to collect, and of varying quality. Thus, the combination of importance and difficulty makes this an exciting area of applied data science.

As with the previous study by Nguyen et al. [48], this paper has outlined the key data challenges and experimented with a range of models to predict S3 emissions. While caution should be taken when comparing results due to differing samples, my findings provide superior predictive performance. However, the main contribution of this work is not in demonstrating model superiority but in illustrating how issues with the underlying data—such as data coverage and quality—impact the model performance.

Through a systematic, experimental approach, I have shown how key choices around the target variable, feature selection, observation sets, and model architecture shape prediction effectiveness. Selecting emission intensities and a core set of features, combined with improving data quality by focusing on material disclosures have proven to be essential for enhancing predictive accuracy.

In addition, the applications in Chapter 6 provide further insights. By using only material observations, the issue of activity exclusion is mitigated, resulting in a more meaningful prediction target and the implementation of the precautionary principle. This represents a novel contribution to the field. The examples in this Chapter also underline the importance of carefully referring to the methodological information published alongside the raw data.

To conclude, this paper has primarily comprised a regression exercise focused on predicting S3 emissions. Alongside Nguyen et al. [48], I find that ML approaches offer limited improvements over simpler methods such as IFM or OLS, especially given the trade-offs in terms of explainability. Nevertheless, careful data work has the potential to improve predictions and account for why models may struggle. In particular, Chapter 6 ties together how the data problems anticipated in Chapter

2 can help explain the results from Chapter 5. Furthermore, in Chapter 7, I have outlined several promising avenues where ML could further address the persistent issues of data quality and coverage. Ultimately, my findings suggest that the future of S3 estimation lies in improving data collection and reporting practices rather than refining model architecture.

Bibliography

[1] Svante Arrhenius. On the influence of carbonic acid in the air upon the temperature of the ground. *Philosophical Magazine and Journal of Science*, 41(251):237–276, 1896. pages 4

[2] Bloomberg. Bloomberg's greenhouse gas emissions estimates model: A summary of challenges and modeling solutions, 2023. Accessed: 2024-08-19. pages 31

[3] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. pages 37

[4] Timo Busch, Matthew Johnson, and Thomas Pioch. Corporate carbon performance data: Quo vadis? *Journal of Industrial Ecology*, 26(1):350–363, 2022. pages 13

[5] Helen Huifen Cai and Mengyao Xia. The driving factors of corporate carbon emissions: an application of the lasso model with survey data. *Environmental Science and Pollution Research*, 30(19):56484–56512, 2023. pages 14

[6] CDP. Cdp supply chain report 2022, 2022. Accessed: 2024-05-22. pages 10

[7] CDP. High quality mandatory disclosure, 2023. Accessed: 2024-05-14. pages 11

[8] CDP. Technical annex iii: Statistical framework. `https://cdn.cdp.net/cdp-production/comfy/cms/files/files/000/008/612/original/Technical_Annex_III_Statistical_Framework_2023.pdf.pdf`, 2023. Accessed: 2024-05-30. pages 14

[9] CDP. Technical note: Scope 3 emissions relevance by sector. Technical report, CDP Worldwide, July 2023. Accessed: 2024-09-30. pages 28

[10] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. *arXiv preprint arXiv:1603.02754*, 2016. pages 37

[11] European Commission. Artificial intelligence act: Article 13. `https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689`, 2024. Accessed: 2024-10-02. pages 2

[12] Robert H. Crawford. Validation of a hybrid life-cycle inventory analysis method. *Journal of Environmental Management*, 88(3):496–506, 2008. pages 8

[13] Luke Cullen, Andrea Marinoni, and Jonathan Cullen. Machine learning for gap-filling in greenhouse gas emissions databases. *Journal of Industrial Ecology*, 28(4):636–647, 2024. pages 58

[14] Antoine Cully, Marek Rei, and Josiah Wang. Lecture notes in introduction to machine learning: Lecture 4, 2023. pages 35

[15] Florence Depoers, Thomas Jeanjean, and Tiphaine. Voluntary disclosure of greenhouse gas emissions: Contrasting the carbon disclosure project and corporate reports. *Journal of Business Ethics*, 134(3):445–461, 2016. pages 10

[16] John Downie and Wendy Stubbs. Evaluation of australian companies' scope 3 greenhouse gas emissions assessments. *Journal of Cleaner Production*, 56(1):156–163, 2013. pages 9

[17] Nils Gustaf Ekholm. On the variations of the climate of the geological and historical past and their causes. *Quarterly Journal of the Royal Meteorological Society*, 27(117):1–62, 1901. pages 4

[18] European Commission. Commission delegated regulation (eu) 2023/2772 of 31 july 2023 supplementing directive 2013/34/eu of the european parliament and of the council as regards sustainability reporting standards. *OJ*, L:1–284, 2023-12-22. pages 6

[19] Japan Financial Services Agency. Publication of draft amendments to the cabinet office order on disclosure of corporate information, etc., 2022. Accessed: 2024-05-14. pages 6

[20] Fluidra. Fluidra sustainability report 2022, 2022. Accessed: 2024-09-24. pages 53

[21] Fluidra. Fluidra annual financial report 2023, 2023. Accessed: 2024-09-24. pages 53

[22] Eunice Foote. Cicumstances affecting the heat of the sun's rays. *The American Journal of Science and Arts*, 22(66):382–383, 1856. pages 4

[23] FTSE Russell. Mind the gaps: Clarifying corporate carbon. Technical report, London Stock Exchange Group, 2022. Accessed: 2024-05-25. pages 13, 22, 45, 46

[24] FTSE Russell. FTSE All-World Index factsheet. `https://research.ftserussell.com/Analytics/Factsheets/Home/DownloadSingleIssue?issueName=AWORLDS&IsManual=False`, 2024. Accessed: 2024-07-31. pages 16

[25] FTSE Russell. Solving the scope 3 conundrum, 2024. Accessed: 2024-05-14. pages 1, 6, 7, 9, 10, 11, 12, 21, 22, 28, 31, 57

[26] Global Reporting Initiative. Gri 305: Emissions 2016, 2016. Accessed: 2024-05-14. pages 5, 10

[27] Bernhard Goldhammer, Christian Busse, and Timo Busch. Estimating corporate carbon footprints with externally available data. *Journal of Industrial Ecology*, 21(5):1165–1179, 2017. pages 8, 14

[28] London Stock Exchange Group. Refinitiv esg scores methodology, 2023. Accessed: 2024-08-14. pages 19

[29] You Han, Achintya Gopal, Liwen Ouyang, and Aaron Key. Estimation of corporate greenhouse gas emissions via machine learning, 2021. pages 14

[30] Frank E. Harrell. *Multivariable Modeling Strategies*, pages 63–102. Springer International Publishing, Cham, 2015. pages 26

[31] Maximilian Hettler and Lorenz Graf-Vlachy. Corporate scope 3 carbon emission reporting as an enabler of supply chain decarbonization: A systematic review and comprehensive research agenda. *Business Strategy and the Environment*, 33(2):263–282, 2024. pages 13

[32] Andreas G. F. Hoepner and Joeri Rogelj. Emissions estimations should embed a precautionary principle. *Nature Climate Change*, 11:638–640, 2021. pages 11, 32, 54

[33] International Financial Reporting Standards Foundation. Ifrs s2 climate-related disclosures, 2023. Accessed: 2024-05-14. pages 5, 10

[34] International Monetary Fund. World economic outlook database, april 2023: Groups and aggregates. `https://www.imf.org/en/Publications/WEO/weo-database/2023/April/groups-and-aggregates`, 2023. Accessed: 2023-08-14. pages 17

[35] IPCC. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, volume In Press. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 2021. pages 4

[36] ISS. Carbon footprint methodology. `https://www.issgovernance.com/file/publications/methodology/Carbon-Footprint-Methodology.pdf`, 2023. Accessed: 2024-05-30. pages 13

[37] Ayush Jain, Manikandan Padmanaban, Jagabondhu Hazra, Shantanu Godbole, and Kommy Weldemariam. Supply chain emission estimation using large language models, 2023. pages 59

[38] Vitali Kalesnik, Marco Wilkens, and Jonas Zink. Do corporate carbon emissions data enable investors to mitigate climate change? *Journal of portfolio management*, 48(10):119–147, 2022. pages 13, 14, 32

[39] Lena Klaaßen and Christian Stoll. Harmonizing corporate carbon footprints. *Nature Communications*, 12(6149), 2021. pages 9, 10, 31

[40] Legal & General Investment Management. Scope 3: Omission impossible, 2023. Accessed: 2024-05-14. pages 10

[41] Lockheed Martin. Carbon strategy and climate-related risk, 2024. Accessed: 2024-09-24. pages 50

[42] Lockheed Martin. Lockheed martin uk carbon reduction plan 2024, 2024. Accessed: 2024-09-24. pages 50

[43] London Stock Exchange Group. Lseg climate data package: Ghg emissions methodology process. `https://www.lseg.com/content/dam/data-analytics/en_us/documents/methodology/lseg-climate-data-package-ghg-emissions-methodology-process.pdf`, 2023. Accessed: 2024-05-30. pages 13

[44] Scott Lundberg and Su-In Lee. A unified approach to interpreting model predictions, 2017. pages 26

[45] Man Group. Carbon emissions: Counting the cost. *Journal of Environmental, Social, and Governance Investing*, 2022. Accessed: 2024-05-25. pages 7

[46] Cristian Mejia and Yuya Kajikawa. Estimating scope 3 greenhouse gas emissions through the shareholder network of publicly traded firms. *Sustainability Science*, 2024. pages 59

[47] Mitsubishi Corporation. Sustainability report 2022, 2022. Accessed: 2024-09-24. pages 52

[48] Quyen Nguyen, Ivan Diaz-Rainey, Adam Kitto, Ben I. McNeil, Nicholas A. Pittman, and Renzhu Zhang. Scope 3 emissions: Data quality and machine learning prediction accuracy. *PLOS Climate*, 2(11):1–31, 11 2023. pages 1, 11, 13, 14, 15, 17, 26, 31, 34, 36, 43, 44, 57, 61

[49] Quyen Nguyen, Ivan Diaz-Rainey, and Duminda Kuruppuarachchi. Predicting corporate carbon footprints for climate finance risk analyses: A machine learning approach. *Energy Economics*, 95:105129, 2021. pages 1, 13, 14, 16, 31, 34, 36

[50] Malgorzata Olesiewicz and Jaakko Kooroshy. Proceed with caution: Challenges in embedding the precautionary principle in paris-aligned benchmarks, Feb 2023. pages 32, 54

[51] Malgorzata Olesiewicz, Jaakko Kooroshy, and Sonja Greven. Navigating the corporate disclosure gap: Modelling of missing not at random carbon data, 2021. pages 11, 14, 21

[52] Georgios Papadopoulos. Discrepancies in corporate ghg emissions data and their impact on firm performance assessment. *SSRN Electronic Journal*, 07 2022. pages 1, 7

[53] Partnership for Carbon Accounting Financials (PCAF). The global ghg accounting and reporting standard for the financial industry. Technical report, 2022. pages 12

[54] Jerry Patchell. Can the implications of the ghg protocol's scope 3 standard be realized? *Journal of Cleaner Production*, 185:941–958, 2018. pages 9, 10

[55] Wouter Poortinga, Lorraine Whitmarsh, Katharine Steentjes, Emily Gray, Sophie Thompson, and Rachel Brisley. Factors and framing effects in support for net zero policies in the united kingdom. *Frontiers in Psychology*, 14, 2023. pages 4

[56] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "Why Should I Trust You?": Explaining the Predictions of Any Classifier, 2016. pages 26

[57] Hannah Ritchie, Pablo Rosado, and Max Roser. Co2 and greenhouse gas emissions. *Our World in Data*, 2023. https://ourworldindata.org/co2-and-greenhouse-gas-emissions. pages 1, 5

[58] John Stanley Sawyer. Man-made carbon dioxide and the "greenhouse" effect. *Nature*, 239(5366):23–26, 1972. pages 4

[59] Securities and Exchange Commission. The enhancement and standardization of climate-related disclosures for investors, 2024. Federal Register, Volume 89, Issue 61, Pages 21668-21921. pages 6, 10

[60] George Serafeim and Gladys Velez Caicedo. Machine learning models for prediction of scope 3 carbon emissions, 2022. Accessed: 2024-05-30. pages 14

[61] Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning important features through propagating activation differences, 2019. pages 26

[62] B. W. Silverman and M. C. Jones. E. fix and j.l. hodges (1951): An important contribution to nonparametric discriminant analysis and density estimation: Commentary on fix and hodges (1951). *International Statistical Review*, 57(3):233–238, 1989. pages 36

[63] Aurel Stenzel and Israel Waichman. Supply-chain data sharing for scope 3 emissions. *npj Climate Action*, 2023. Accessed: 2024-05-25. pages 11

[64] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *International conference on machine learning*, pages 3319–3328. PMLR, 2017. pages 26

[65] Laurens Swinkels and Thijs Markwat. Corporate carbon emissions data for equity and bond portfolios. *Managerial Finance*, 50(1):118–139, 2024. pages 13

[66] Kevin Tang, Gireesh Shrimali, and Christophe Christiaen. Data quality considerations for estimating financed emissions. Technical report, Smith School of Enterprise and the Environment, University of Oxford, January 2023. Discussion Paper. pages 13

[67] Task Force on Climate-related Financial Disclosures. Implementing the recommendations of the task force on climate-related financial disclosures, 2021. Accessed: 2024-05-14. pages 5, 10

[68] The Royal Society and the US National Academy of Sciences. Climate change evidence and causes: Update 2020. Technical report, 2020. pages 4

[69] UK Department for Environment, Food & Rural Affairs. Environmental reporting guidelines: Including streamlined energy and carbon reporting guidance, 2019. Accessed: 2024-05-31. pages 6

[70] UK Government. The companies act 2006 (strategic report and directors' report) regulations 2013, 2013. pages 6

[71] UK Government. A roadmap towards mandatory climate-related disclosures, 2020. Accessed: 2024-05-14. pages 7

[72] UNFCCC. The Paris Agreement, 2020. pages 4

[73] United Nations. Kyoto protocol to the united nations framework convention on climate change. *United Nations Treaty Collection*, 2303:162, 1997. pages 4

[74] United Nations Framework Convention on Climate Change. The Paris Agreement. Accessed: 2024-05-30. pages 1

[75] World Business Council for Sustainable Development and World Resources Institute. *The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard*. World Business Council for Sustainable Development, 2004. pages 1, 5

[76] World Business Council for Sustainable Development and World Resources Institute. *Technical Guidance for Calculating Scope 3 Emissions*. World Business Council for Sustainable Development, 2013. pages 6, 7, 8, 12

[77] Davis A Young. History of the collapse of "flood geology" and a young earth, 2007. Accessed: 2024-05-14. pages 4

[78] Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005. pages 34

Appendix A

Scopes regression

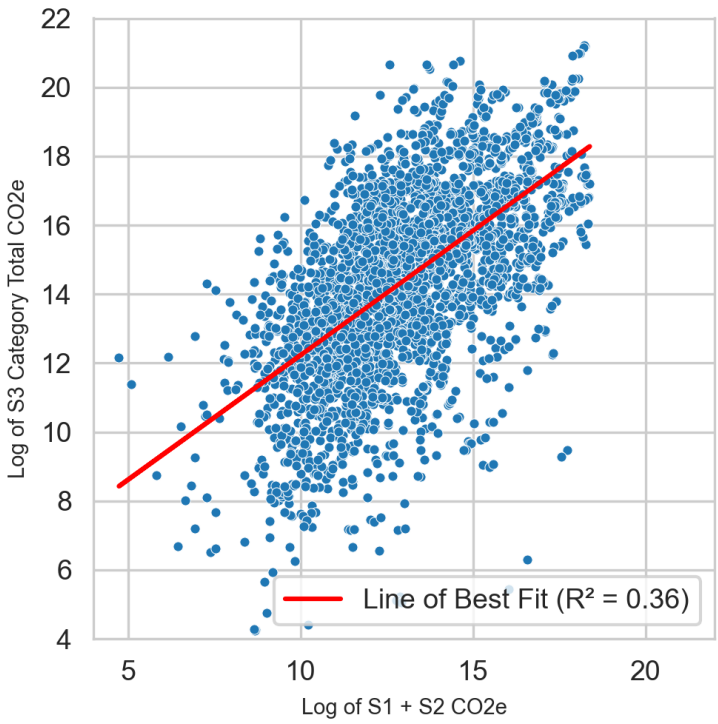


Figure A.1: Scopes 1 and 2 vs Scope 3 emissions

Appendix B

Material sectors

Table B.1: Material Scope 3 GHG emissions categories by sector

TRBC Code	TRBC Name	Most material S3 categories
5010	Energy - Fossil Fuels	1, 11
5020	Renewable Energy	1, 11
5110	Chemicals	1, 11
5120	Mineral Resources	1, 11
5130	Applied Resources	1, 12
5210	Industrial Goods	1, 11
5220	Industrial and Commercial Services	1, 11
5240	Transportation	1, 3
5310	Automobiles and Auto Parts	1, 11
5320	Cyclical Consumer Products	1, 11
5330	Cyclical Consumer Services	1, 14
5340	Retailers	1, 11
5410	Food and Beverages	1, 4
5420	Personal and Household Products and Services	1, 11
5430	Food and Drug Retailing	1, 11
5440	Consumer Good Conglomerates	1, 11
5610	Healthcare Services and Equipment	1, 11
5620	Pharmaceuticals and Medical Research	1, 2
5710	Technical Equipment	1, 11
5720	Software and IT Services	1, 11
5730	Financial Technology and Infrastructure	1, 11
5740	Telecommunications Services	1, 2
5910	Utilities	3, 11
6010	Real Estate	2, 13

Appendix C

Sample summary statistics

Table C.1: Summary description of numeric variables in the full sample

Feature	Types	Total	Missing	Available	Missing_Percent	Unique	Min	Max	Mean	Median	Std Error	Kurtosis	Skewness	Quantile5	Quantile95	Quantile25	Quantile75	Quantile10	Quantile90
mcap	float64	10798	0	10798	0	10768	128472959	2.90164E+12	27981082264	10007886349	90791948420	358	16	1822541548	98057425494	4732325367	24105221137	2570952733	53967531542
revenue	float64	10798	0	10798	0	10733	34979642	6.03653E+11	17110002296	6671618551	35885785321	67	7	728535124	65758200711	2680301725	16278293865	1224415476	39604030524
ebit	float64	10798	0	10798	0	10610	-17642624021	3.0476E+11	2021894642	715178929	6449325801	740	21	39067569	7807542162	303925519	1746352014	119751063	4313300000
ebitda	float64	10798	0	10798	0	10646	-10485816305	3.2913E+11	3092841968	1133788246	8127442794	404	15	129901083	12145882917	492152276	2739298784	230601630	6479348727
net_cash_flow	float64	10798	0	10798	0	10515	-1792287778	38010000000	2265177263	832036975	4664781347	28	5	29798485	9264104224	338804957	2113440134	132204377	5112131779
assets	float64	10798	0	10798	0	10741	74211519	6.63541E+11	27389642043	11347265929	51156734723	34	5	1555725283	1.0872E+11	4862185062	26940914779	2351645867	61413675821
receivables	float64	10798	12	10786	0	10573	-120524654	2.06076E+11	3766062592	1159144840	10767457231	136	10	94282603	13250881338	467504832	3004731932	180807105	7939315965
gross_ppe	float64	10798	141	10657	0.01	10271	536860	6.47768E+11	16663711407	4916904923	40628206420	64	7	297049182	63281496031	1583485083	14392834779	589732739	38146000000
net_ppe	float64	10798	14	10784	0	10652	56022	4.66349E+11	8274557234	2443857615	20238826743	96	8	132594969	34048600996	781267103	7027353177	286647965	19867279366
capex	float64	10798	0	10798	0	10428	212385	67882000000	1233582794	409820578	3059162139	85	8	36999039	4608948961	151404535	1034795150	66150245	2634263422
intangible_assets	float64	10798	0	10798	0	10395	3952	3.10197E+11	5397328780	1076039631	14815815604	93	8	10891843	23717987912	168884165	4257936079	35609813	12711201346
lt_debt	float64	10798	39	10759	0	10493	2639	1.6625E+11	6502468540	2222507827	13258088805	36	5	12904092	28206100000	579289141	6531712362	60272443	15319123538
policy_emissions_score	float64	10798	2	10796	0	354	0	97	63	65	17	7	-3	0	81	60	71	55	77
target_emissions_score	float64	10798	80	10718	0.01	341	0	96	58	76	36	-1	-1	0	91	0	83	0	88
s1_co2e	float64	10798	0	10798	0	9699	0	257096000	3173737	99434	12244063	98	8	1431	16493200	14834	713148	3471	6459976
s2_co2e	float64	10798	0	10798	0	9493	0	45010000	693575	176486	1969251	166	10	6921	2940277	47000	547354	13945	1611236
s1_and_s2_co2e	float64	10798	0	10798	0	10205	114	272818700	3867313	360574	13437088	95	8	13191	18426560	88603	1641355	26200	7947619
s3_co2e	float64	10798	3943	6855	0.37	6634	0	5142900000	24250181	1211810	112587719	675	19	3529	108042100	103670	9029225	12895	42671800
s3_upstream	float64	10798	7799	2999	0.72	2949	46	360900000	5538481	1305518	18468908	156	11	12964	20029141	229623	4742469	37032	11615731
s3_downstream	float64	10798	8980	1818	0.83	1785	127	1351000000	36149937	2067000	118067357	50	6	23700	197445452	312567	17269010	56988	77171860
s3_cat_total	float64	10798	7602	3196	0.7	3170	46	1637000000	25760479	2057960	101337559	107	9	15735	107980000	352344	10987028	49364	44152049
s3_purchased_goods_cat1	float64	10798	8334	2464	0.77	2380	0	190000000	4513343	1184780	13568036	91	9	11931	16552950	243900	3961532	51010	8807234
s3_capital_goods_cat2	float64	10798	8919	1879	0.83	1650	0	16724600	388769	112142	969807	85	8	881	1594266	31000	329000	6767	957606
s3_fuel_energy_cat3	float64	10798	8450	2348	0.78	2033	0	360900000	1404960	56617	10393704	646	22	1396	4181750	11050	247000	3209	1287000
s3_transportation_cat4	float64	10798	8580	2218	0.79	1938	0	30180000	520609	80130	1939700	119	10	624	1900000	14596	364891	2387	923100
s3_waste_cat5	float64	10798	8563	2235	0.79	1691	0	13394000	46166	6200	315230	1444	35	54	163751	1055	25000	194	80299
s3_business_travel_cat6	float64	10798	8199	2599	0.76	1923	0	1000000	21889	4684	57942	76	7	121	94078	1222	17698	362	53507
s3_employee_commuting_cat7	float64	10798	8570	2228	0.79	1699	0	3500000	49899	12000	169538	244	13	396	193055	3471	38487	1000	110300
s3_leased_assets_cat8	float64	10798	10011	787	0.93	482	0	3400000	42858	1051	201871	119	9	0	120500	0	11386	0	50278
s3_distribution_cat9	float64	10798	9837	961	0.91	770	0	19900000	536314	50100	1847069	52	7	0	2400000	6920	269000	637	929150
s3_processing_products_cat10	float64	10798	10278	520	0.95	357	0	549000000	8100253	58033	57698544	81	9	0	13014000	29	769078	0	6146437
s3_use_of_sold_cat11	float64	10798	9302	1496	0.86	1434	0	1351000000	39409705	2502000	123859414	48	6	4598	231478250	242763	21205925	22217	82665000
s3_EOL_treatment_cat12	float64	10798	9688	1110	0.9	955	0	127600000	1102599	42550	6415409	248	14	48	3928000	3206	364479	300	1663000
s3_leased_assets_cat13	float64	10798	10235	563	0.95	379	0	10619400	240024	5167	931162	60	7	0	1044382	76	67298	0	399340
s3_franchises_cat14	float64	10798	10455	343	0.97	156	0	7430620	289080	5	1038747	24	5	0	1950539	0	14945	0	240731
s3_investments_cat15	float64	10798	10217	581	0.95	389	0	35286000	994776	28000	4255021	43	6	0	3304000	106	189665	0	1250460

Details of the sample's construction are given in Section 3.3.

Appendix D

Model hyperparameters

D.1 Quantile regression

Table D.1: Hyperparameters of the Quantile Regression model in scikit-learn.

Parameter	Description	Choice Set
`quantile`	The quantile to be predicted, which must be between 0 and 1 (exclusive). For example, 0.5 represents the median regression, 0.9 represents the 90th percentile, etc.	{0.1, 0.25, 0.5, 0.75, 0.9}
`alpha`	Regularisation strength; must be a non-negative float. Higher values imply stronger regularization. In practice, this controls the trade-off between bias and variance. Note that this applies if the quantile regression model is regularized.	{0.01, 0.1, 0.5, 1.0, 10.0}
`solver`	Algorithm used for optimization. The choice of solver can affect the speed and accuracy of the model fitting process.	{'highs-ds', 'highs-ipm', 'highs'}
`fit_intercept`	Boolean indicating whether or not to fit the intercept. If `True`, the model will calculate the intercept. If `False`, no intercept will be used in the model.	{True, False}

D.2 Elastic Net

Table D.2: Hyperparameters of the Elastic Net model in scikit-learn.

Parameter	Description	Choice Set
`alpha`	Regularization strength; must be a non-negative float. A higher value of `alpha` increases the amount of regularization, making the model more biased but potentially reducing variance. When `alpha` is set to 0, no regularization is applied, making the model equivalent to ordinary least squares regression.	{1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1.0, 2.0, 5.0, 10.0, 100.0, 1000.0}
`l1_ratio`	The mixing parameter that determines the balance between L1 (Lasso) and L2 (Ridge) regularization. `l1_ratio` is a float between 0 and 1. When `l1_ratio=1`, the penalty is purely L1 (Lasso). When `l1_ratio=0`, the penalty is purely L2 (Ridge). Values between 0 and 1 provide a combination of both penalties.	{0.0, 0.05, 0.1, . . . , 0.95}
`max_iter`	The maximum number of iterations for the optimization algorithm to converge. If the algorithm does not converge within this number of iterations, it will stop. Increasing `max_iter` may help achieve convergence, especially with large datasets or when strong regularization is applied.	{10000}

D.3 Multilayer perceptron

Table D.3: Hyperparameters of the MLPRegressor model in scikit-learn.

Parameter	Description	Choice Set
`hidden_layer_sizes`	A tuple representing the number of neurons in each hidden layer.	{(10,), (50,), (100,), (10,10), (50,50)}
`activation`	The activation function for the hidden layer. Options include `identity`, `logistic`, `tanh`, and `relu`.	{'tanh', 'relu'}
`alpha`	L2 penalty (regularization term) parameter. This helps prevent overfitting by adding a penalty to the loss function.	{1e-4, 1e-3, 1e-1}
`learning_rate`	Controls the learning rate schedule for weight updates. Options include `constant`, `invscaling`, and `adaptive`.	{'constant', 'invscaling', 'adaptive'}
`learning_rate_init`	The initial learning rate used. It controls the step size at each iteration while moving toward a minimum of the loss function.	{0.1, 0.3, 0.5, 0.7, 0.9, 1.1}
`max_iter`	The maximum number of iterations for the solver to converge. If the model does not converge within this number of iterations, the training stops.	{1000, 1500, 2000}
`early_stopping`	Whether to use early stopping to terminate training when validation score is not improving.	{True, False}

D.4 K Nearest-Neighbours

Table D.4: Hyperparameters of the KNeighborsRegressor model in scikit-learn.

Parameter	Description	Choice Set
`n_neighbors`	The number of neighbors to use for making predictions. This controls how many nearest neighbors are averaged to make a prediction.	{5, 10, 15, . . . , 500}
`weights`	The weighting function used to influence the prediction from the neighbors. Options include: • `uniform`: All neighbors contribute equally to the prediction. • `distance`: Closer neighbors contribute more to the prediction (weights are inversely proportional to distance).	{'uniform', 'distance'}
`algorithm`	The algorithm used to compute the nearest neighbors. Options include: • `auto`: Automatically chooses the best algorithm based on the values passed to `fit` method. • `ball_tree`: Uses the BallTree algorithm. • `kd_tree`: Uses the KDTree algorithm. • `brute`: Uses brute-force search.	{'auto', 'ball_tree', 'kd_tree', 'brute'}
`p`	Power parameter for the Minkowski distance metric.	{3, 4}
`metric`	The distance metric to use for the tree.	{'minkowski', 'euclidean', 'manhattan'}

D.5 XGBoost

Table D.5: Hyperparameters of the XGBoost model in scikit-learn.

Parameter	Description	Choice Set
`max_depth`	The maximum depth of a tree. Increasing this value makes the model more complex and more likely to overfit.	{6, 7, 8, 9, 10, 11}
`min_child_weight`	Minimum sum of instance weight (hessian) needed in a child. This parameter controls the complexity of the model by preventing the creation of nodes that only contain a few samples. A higher value prevents overfitting by making the algorithm more conservative.	{1, 2, 3, 4, 5, 6, 7, 8}
`eta`	The learning rate, also known as `eta`. It scales the contribution of each tree by a factor (between 0 and 1) to prevent overfitting. Smaller values make the model more robust but require more trees.	{0.00001, 0.0001, 0.005, 0.001, 0.05, 0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1}
`lambda`	L2 regularization term on weights. This helps to avoid overfitting by penalizing large coefficients.	{0.00001, 0.0001, 0.005, 0.001, 0.05, 0.01, 0.1, 0.2, 0.3, 0.5, 1.0}
`subsample`	The fraction of samples to be used for fitting individual trees. This parameter can be used to prevent overfitting by sampling a random subset of data. The default is `1` (using all data).	{0.5, 0.6, 0.7, 0.8, 0.9, 1.0}
`colsample_bytree`	The fraction of features (columns) to be used when constructing each tree. Like `subsample`, it helps prevent overfitting by adding randomness.	{0.5, 0.6, 0.7, 0.8, 0.9, 1.0}
`n_estimators`	The number of boosting rounds (i.e., number of trees to fit). More boosting rounds can improve accuracy but increase training time.	Optimised using early stopping over folds, up to 1000.
`scale_pos_weight`	Balancing of positive and negative weights, useful for unbalanced classes. It can make the algorithm converge faster and perform better on imbalanced datasets.	Not specified (use default)
`booster`	The type of booster to use. Options include `'gbtree'`, `'gblinear'`, and `'dart'`. `'gbtree'` is used for tree-based models and is the default.	Not specified (use default)

D.6 Random Forest

Table D.6: Hyperparameters of the RandomForestRegressor model in scikit-learn.

Parameter	Description	Choice Set
`n_estimators`	The number of trees in the forest. Increasing the number of trees generally improves performance but also increases computational cost.	{100, 200, 300}
`max_depth`	The maximum depth of each tree. Limiting the depth can prevent overfitting. If `None`, nodes are expanded until all leaves are pure or until all leaves contain fewer than `min_samples_split` samples.	{3, 5, 10}
`min_samples_split`	The minimum number of samples required to split an internal node. Higher values prevent the model from learning overly specific patterns, which can reduce overfitting.	{2, 10}
`min_samples_leaf`	The minimum number of samples that must be present in a leaf node. Setting this parameter to a higher value can smooth the model and prevent it from fitting noise in the data.	{2, 5}
`max_features`	The number of features to consider when looking for the best split. Options include auto, sqrt, and log2.	{'sqrt', 'log2'}
`bootstrap`	Whether to use bootstrap samples when building trees. If `True`, each tree is built from a random subset of the data with replacement.	{True, False}
`criterion`	The function to measure the quality of a split. Options include mean squared error 'mse', mean absolute error 'mae', and poisson.	{'absolute_error'}